PORTFOLIO

Industrial Design

01

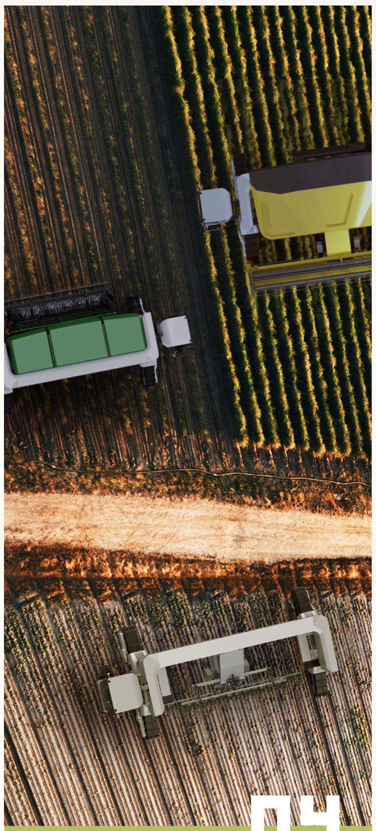
Game Design

02

Furniture Design

03

Product Design



04

Structure Design

01

SERENDIPITY AT THE MUSEUM

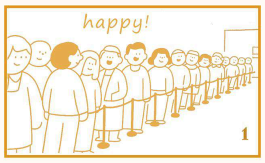
See It in Action

Scan te OR code toexperience the magicof AR

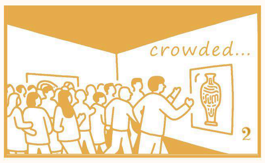
Collaborate with museums to develop an interactive visualnovel, blending storytelling and AR effects for popularscience, Take Tianiin museum as the beginning.

BACKGROUND

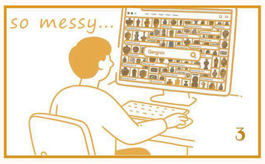
PROBLEM IDENTIFICATION



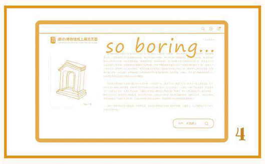
Wait for visiting.



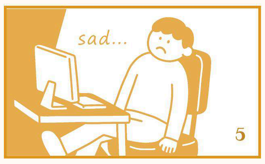
Too crowded to deeply learn.



Tired of searching online.



Website's science content is somewhat dull.

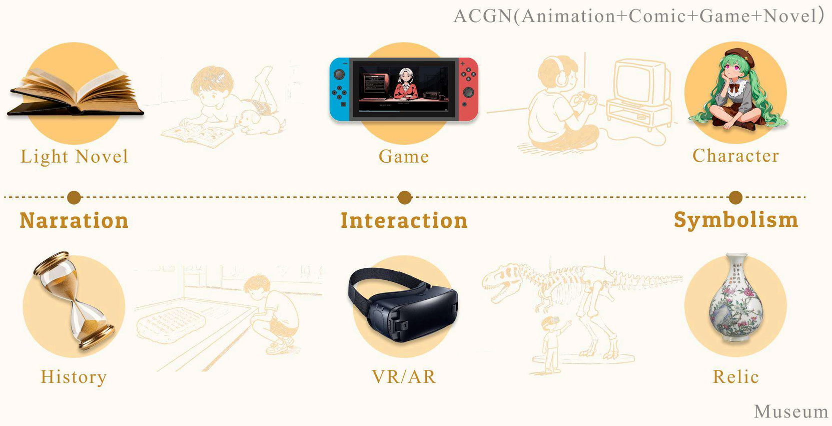


Feel deeply helpless about this.

Maybe there will be a better way ?

Museums offer rich history, but the experience is often overwhelming and disconnected.

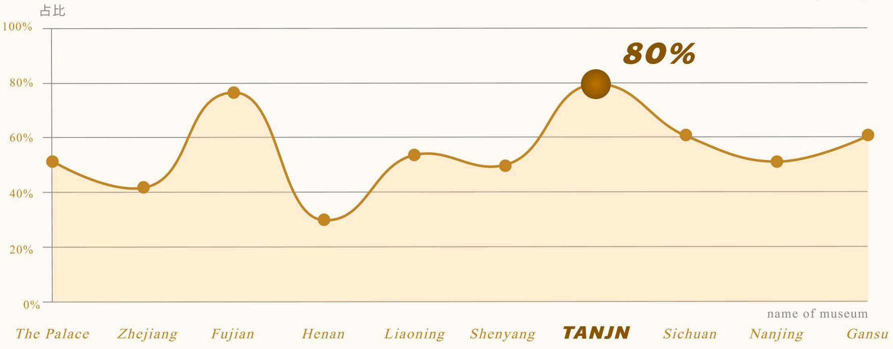
INSPIRATION FROM ACGN


ACGN(Animation+Comic+Game+Novel)
Light Novel
Game
Character
Narration
Interaction
Symbolism
History
VR/AR
Relic
Museum

ACGN and museum popularization share similar features.Some of ACGN's expressive forms can be borrowed.

WHY IS THE TIANJIN MUSEUM

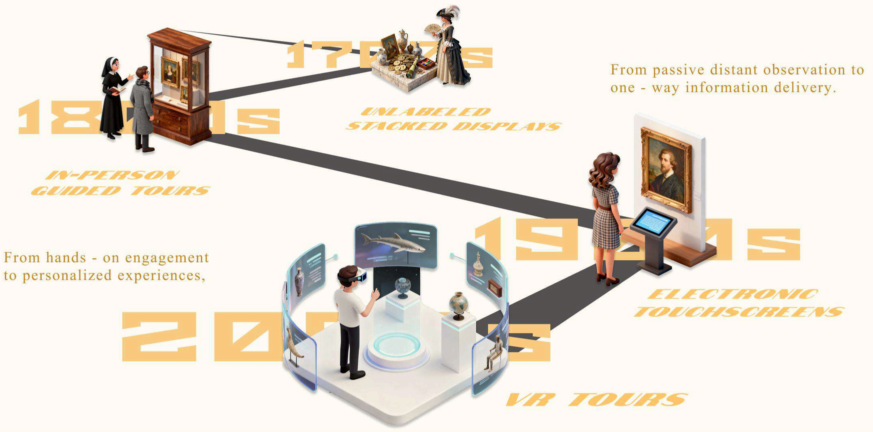
Compiled from the Museum's Public Information (2023)



Proportion of Precious Cultural Relics

Relics vary; some collections overload visitors, others engage

MUSEUM INTERACTION DEVELOPMENT


17 s
UNLABELED STACKED DISPLAYS
18 s
IN-PERSON GUIDED TOURS
19 s
ELECTRONIC TOUCHSCREENS
20 s
VR TOURS

From passive distant observation to one - way information delivery.

From hands - on engagement to personalized experiences,

Moving from static info to dynamic, engaging, personalized

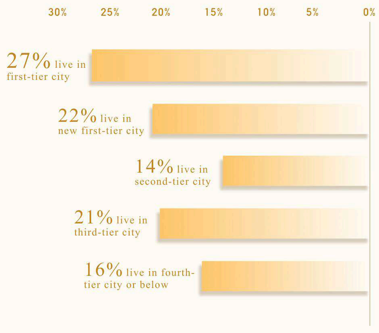
effectively. Curated narratives improve guidance.

journeys with AR and storytelling.

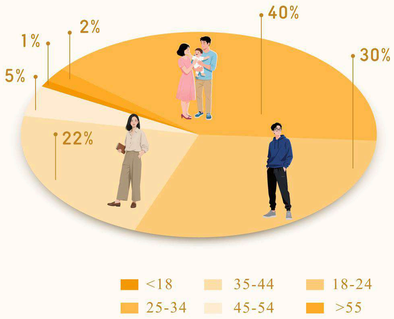
RESEARCH

PRELIMINARY INVESTIGATION

People mainly in big cities.



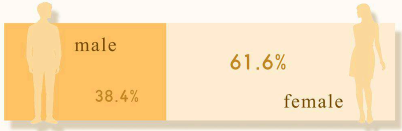
In big cities, Majority is young generation



1.Urban Distribution
Use the customary expressions for classifying Chinese cities

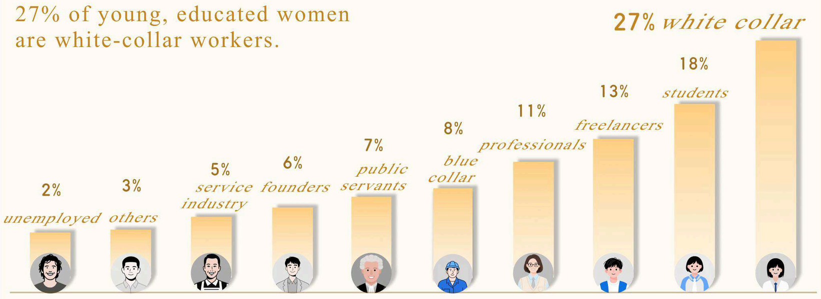
2.Age Distribution

Educated women outnumber men in large cities.



3.Sexual Distribution

27% of young, educated women are white-collar workers.



4.Career Distribution

Core Audience: Young, educated urban women seeking meaningful, story-driven cultural content.

PERSONA

Age:28
Job:freelancer
Address: Beijing

Introduction:

She love to treasure content depth and evidence - based research, and pursues layered knowledge acquisition.

Pain Point:

Information is disorganized and lacks systemization

Needs:

In-depth knowledge, efficient information retrieval

Age:25
Job:white collar
Address: Tianjin

Introduction:

She likes content with a plot, values the sense of participation, and focuses on the connection between popular science and life.

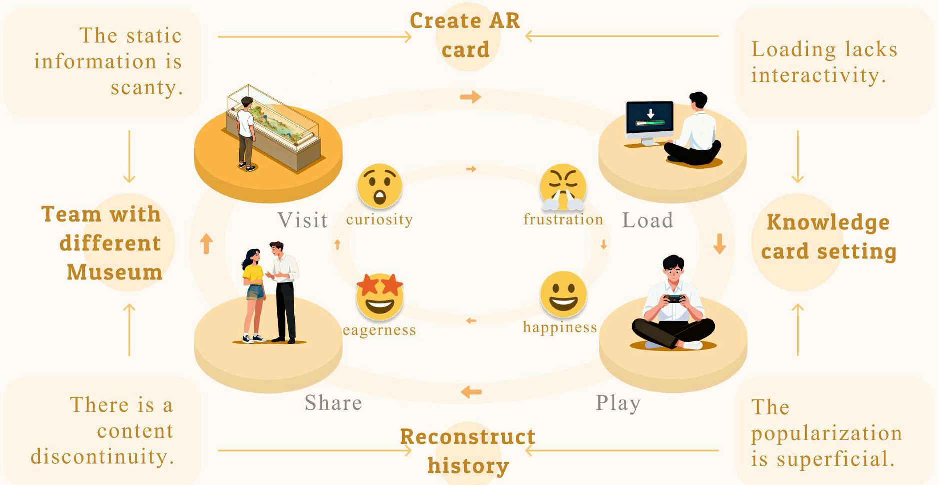
Pain Point:

The information is too esoteric and tedious

Needs:

A more audience - resonant approach to science storytelling

USER JOURNEY MAP


curiosity
frustration
eagerness
happiness

The static information is scanty.

Create AR card

Loading lacks interactivity.

Team with different Museum

Visit

Load

Knowledge card setting

Share

Play

There is a content discontinuity.

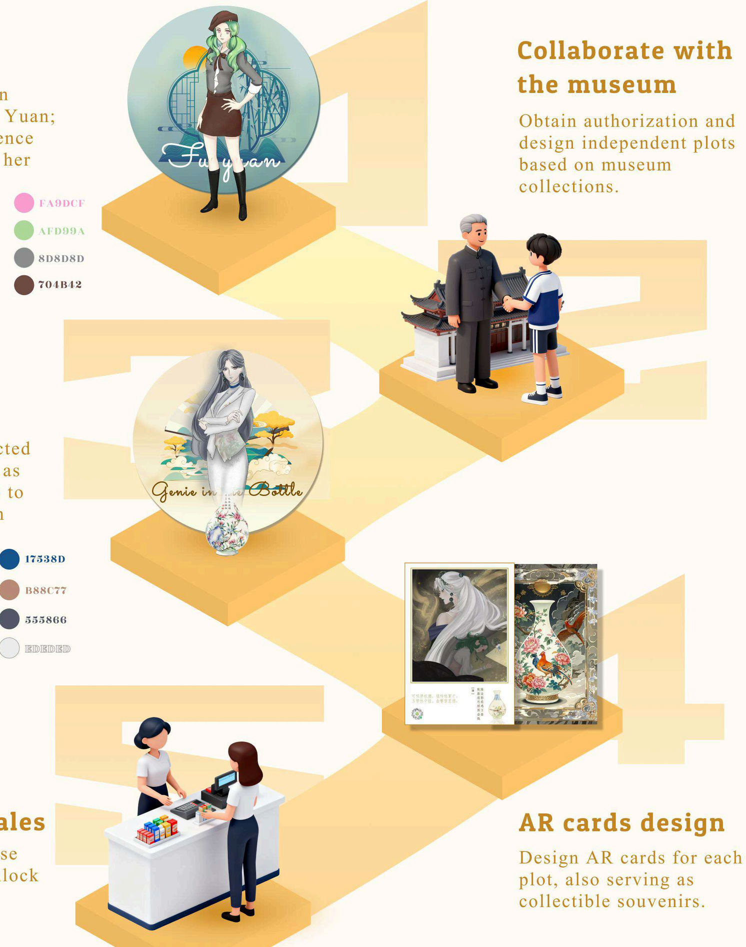
Reconstruct history

The popularization is superficial.

Balance logical knowledge flow with emotionally engaging scenarios to meet diverse user needs.

DESIGN PROCESS

THE OVERALL MODE OPERATION


Fu yuan
FA9DCF
AFD99A
8D8D8D
704B42
Genie in the Bottle
17538D
B88C77
555866
EDEDED

Main plot

Create the main protagonist Fu Yuan; players experience the game from her perspective.

Collaborate with the museum

Obtain authorization and design independent plots based on museum collections.

Side plot

Personify selected museum relics as side characters to spread museum knowledge.

Product sales

Players purchase AR cards to unlock related plots.

AR cards design

Design AR cards for each plot, also serving as collectible souvenirs.

BUSINESS MODEL CANVAS

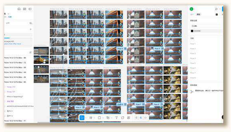
Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
1.Museums 2.game platforms 3.AR/merch manufacturers 4.offline retailers.	1.Co-develop stories 2.design games/AR cards/merch 3.manage sales channels & community	1. Combine relics with interactive storytelling to make learning fun 2. AR cards: physical + digital (unlock exclusive stories, collectible) 3. Relic/character merch for personalized needs	1.Online communities (story/collection chats) 2.offline museum shows 3.regular game updates	1.Cultural heritage lovers 2.young gamers 3.creative goods buyers (collection/interaction needs)
	Key Resources		Channels	
	1.Museum IP 2.content team 3.supply chain 4.sales network		1.Online: game platforms, official site, e-commerce (games/AR cards/merch) 2.Offline: museum stores, creative shops.	

Cost Structure	Revenue Streams
1.Content dev (games/AR/art) 2.collaboration (licensing/joint dev) 3.production/logistics 4.marketing	1.Game sales 2.AR card sales (story unlocks) 3.merch sales 4.museum profit sharing

Combines cultural heritage IP with AR storytelling for sustainable, engaging cultural experiences.

TECHNOLOGY IMPLEMENTATION

Prototyping (Figma)



1.interactive link
2.animation setting
3.music integration

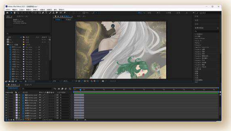
Asset Preparation (Ps)

Digitally design the card face and organize layers properly

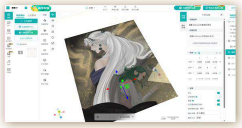
2D Animation (Ae)

AR Implementation (Kivicube)

We employ the plot setting of Side Arc-Dominant Structure with a Connective Main Plo to string together multiple plots designed for the museum.

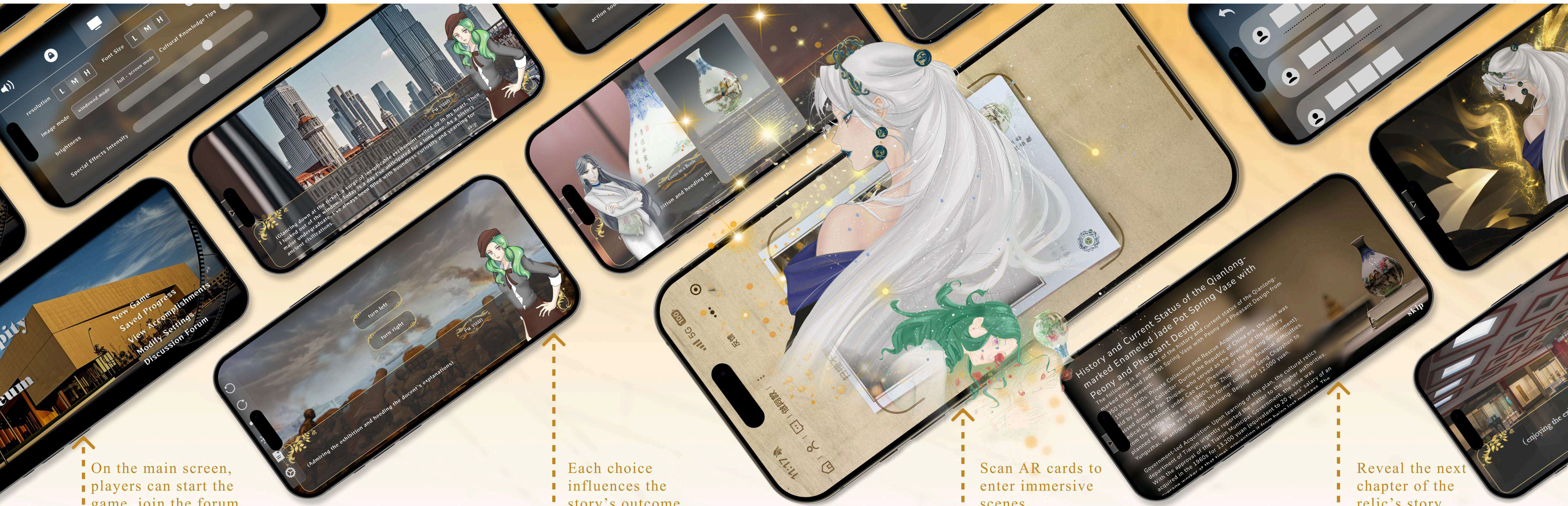


1.import vector graphics
2.layer management
3.keyframe design
4.Interaction preview



1.create project
2.visual editing
3.configure interaction logic
4.preview testing

https://www.kivicube.com/

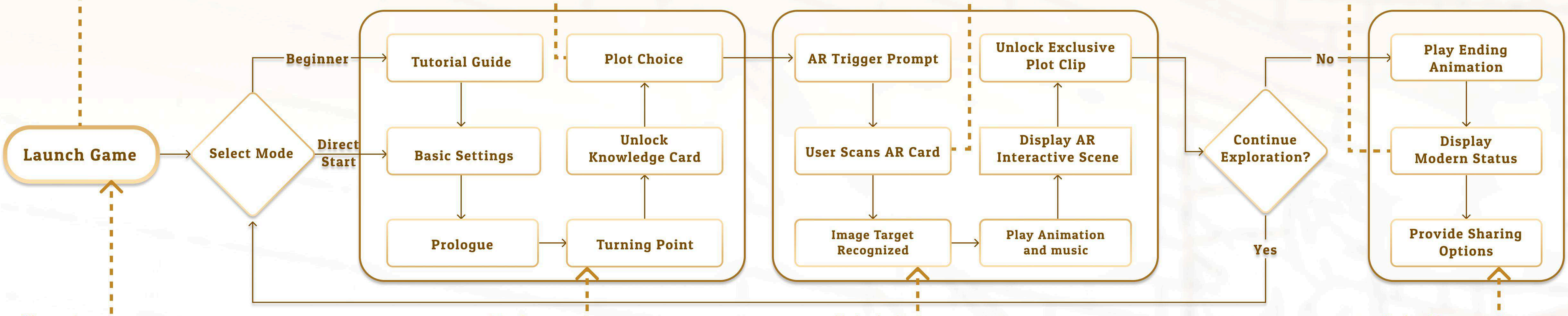


On the main screen, players can start the game, join the forum, or adjust settings.

Each choice influences the story's outcome.

Scan AR cards to enter immersive scenes.

Reveal the next chapter of the relic's story.



Launch Game

Select Mode

Beginner

Direct Start

Tutorial Guide

Basic Settings

Prologue

Turning Point

Unlock Knowledge Card

Plot Choice

AR Trigger Prompt

User Scans AR Card

Image Target Recognized

Play Animation and music

Display AR Interactive Scene

Unlock Exclusive Plot Clip

Continue Exploration?

No

Yes

Play Ending Animation

Display Modern Status

Provide Sharing Options

Play calm background music to set the tone.

Music intensifies at story turning points.

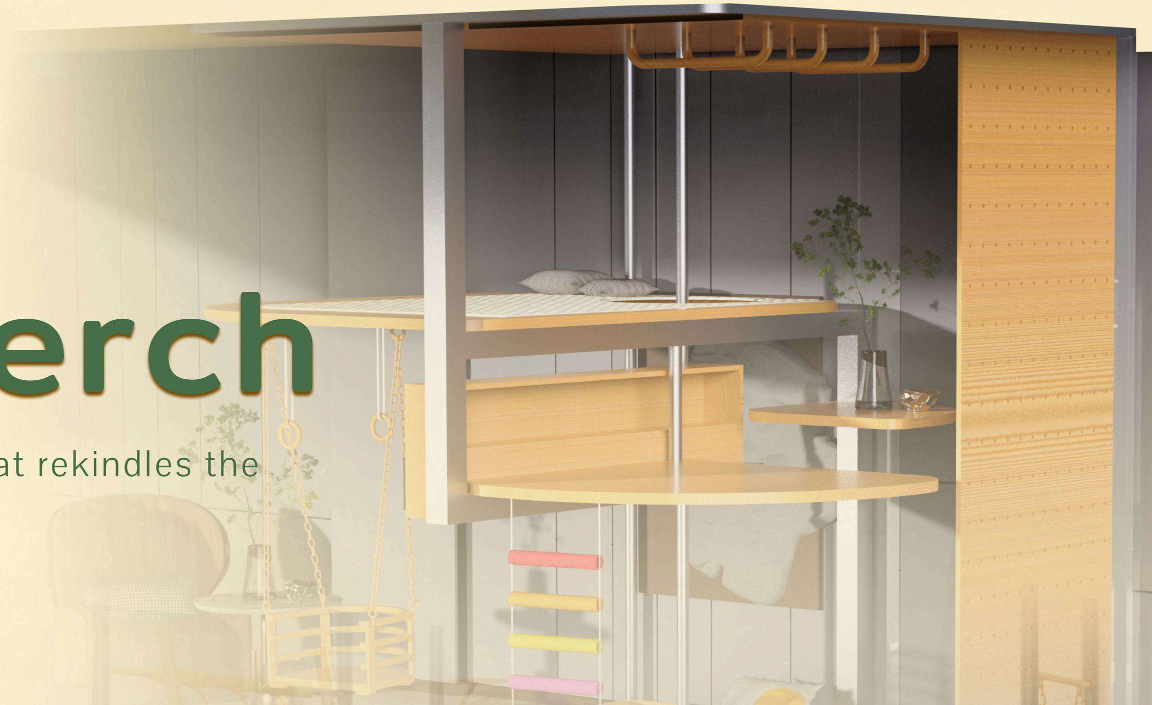
Unlock the AR card at the story's climax.

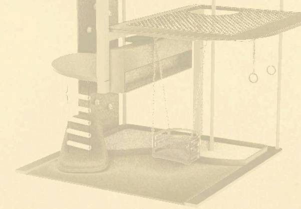
End with lingering music that echoes the emotion.

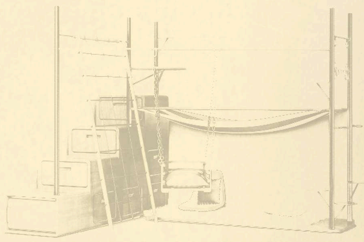
02

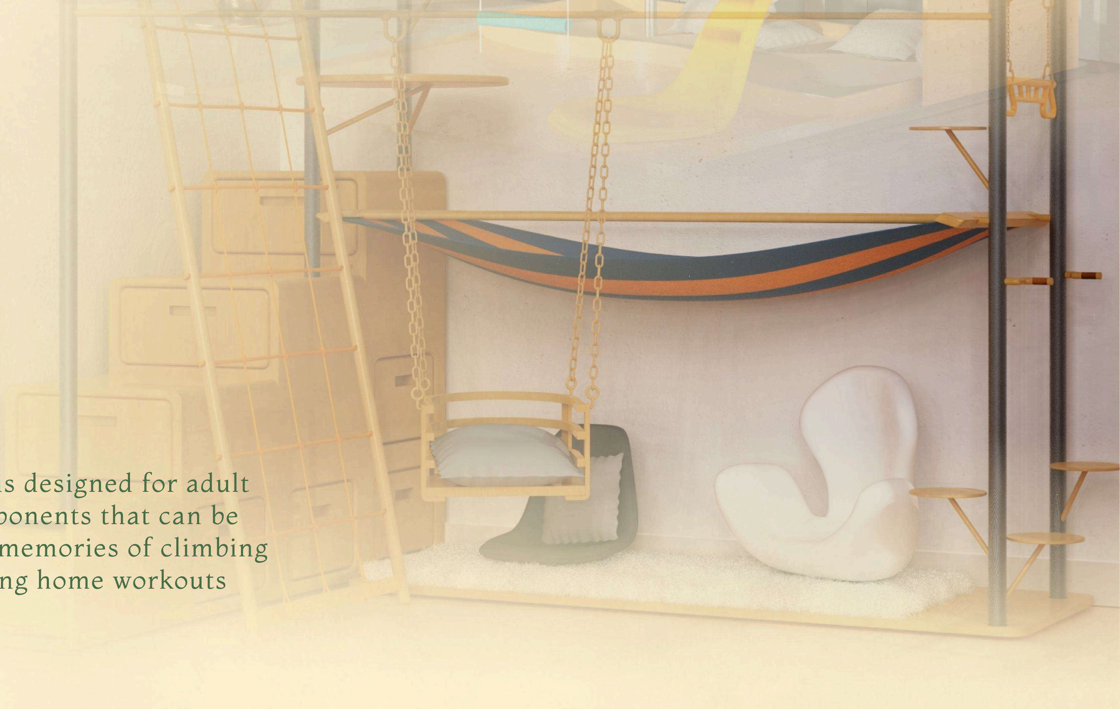
Wildling Perch

An anthropomorphic climbing frame that rekindles the carefree spirit of childhood at home.





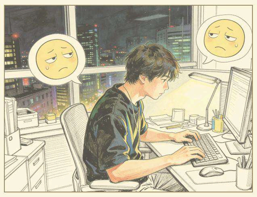




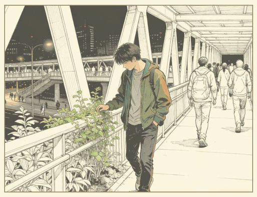
This home climbing frame, inspired by cat climbing structures, is designed for adult use. It features multiple modular frames and customizable components that can be assembled through an app. The design aims to evoke childhood memories of climbing and the unrestrained joy of nature, while also promoting engaging home workouts and alleviating physical strain caused by prolonged office work.

BACKGROUND

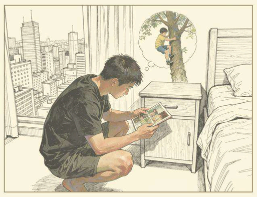
THE PROBLEM OF URBANITES



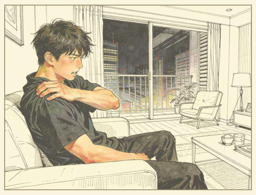
1.Quiet nights — everyone's still working.



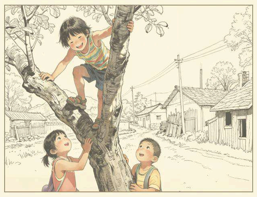
2.Feeling lost, unsure what to chase.



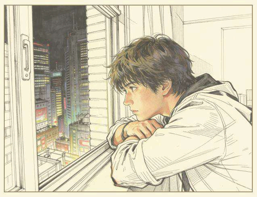
3.Once had a carefree childhood.



4.Illness keeps them awake.



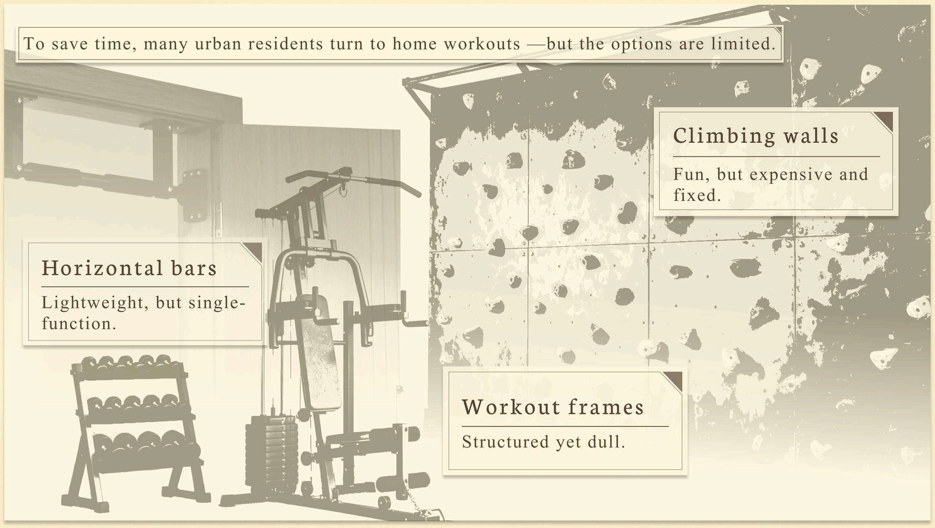
5.Used to love climbing.



6.Now, it's just a distant memory.

Urban youth face two main struggles — health and homesickness.

EXISTING HOME FITNESS SOLUTIONS



To save time, many urban residents turn to home workouts —but the options are limited.

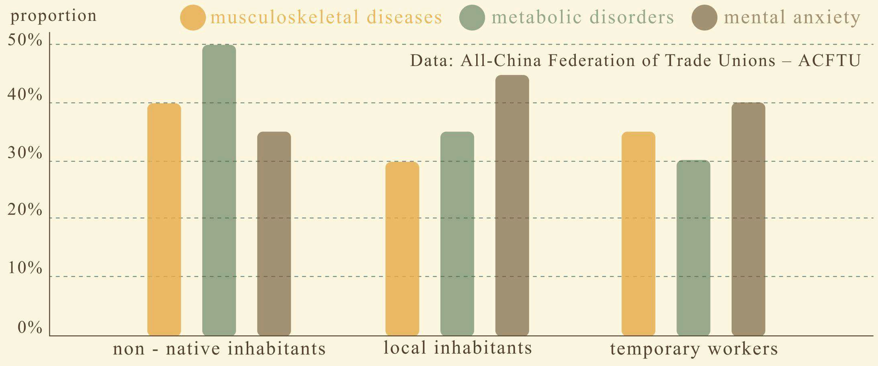
Climbing walls
Fun, but expensive and fixed.

Horizontal bars
Lightweight, but single-function.

Workout frames
Structured yet dull.

Why not redesign and assemble them into something new?

WHAT HEALTH PROBLEMS DO PEOPLE FACE



Data: All-China Federation of Trade Unions – ACFTU

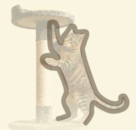
Chronic illness and mental stress are the main afflictions.

INSPIRATION DERIVED FROM THE CAT TREE

Demand for Comfort

Cat:
Finds a cozy resting spot.

Human:
Creates a relaxation zone.



Demand for Release

Cat:
Climbs and plays freely.

Human:
recreational exercise

Demand for Health

Cat:
Sharpen claws.

Human:
Strengthens the body.

Demand for Boundary

Cat:
Seek solitude.

Human:
Desires personal space.

Why not design a human climbing frame,just like a cat tree?

POSITIONING

USER CHARACTERISTICS AND BEHAVIORS

OPERATION SYSTEM

Age: 32 *Basic*
Job:Engineer
Residence:120㎡ industrial-style apartment in Hangzhou
Chen Mo

Introduction

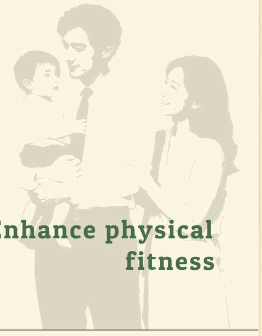
Married with a 2-year-old child.
Grew up in the countryside, often climbing camphor trees with friends.

Needs

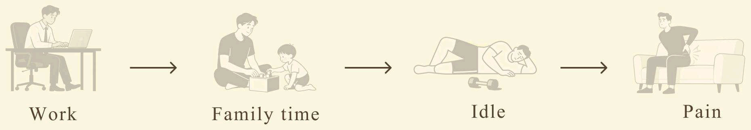
Wants a compact home exercise device that relieves neck and back pain, runs quietly, and saves space.

Pain point

Limited time
Noise concerns
Neck and lumbar strain
Idle equipment



Enhance physical fitness


Work
Family time
Idle
Pain

Behaviors

Age: 28 *Basic*
Job:Interior Planner
Residence:68㎡ Japanese-style apartment in Shanghai
Lin Xiao

Introduction

Single, grew up in Yancheng, Jiangsu.
Used to climb phoenix trees at school with classmates.

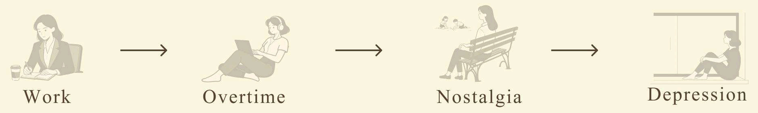
Needs

Wants a climbing setup that brings back childhood memories, matches minimalist décor, and enhances mood.

Pain point

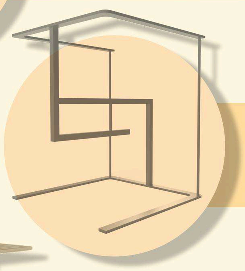
Emotional loneliness
Aesthetic mismatch
Boring workouts
Lack of nostalgia

Evoke childhood memories

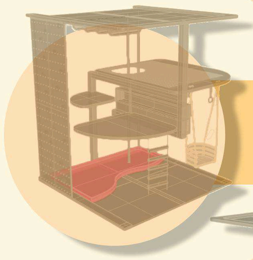

Work
Overtime
Nostalgia
Depression

Behaviors

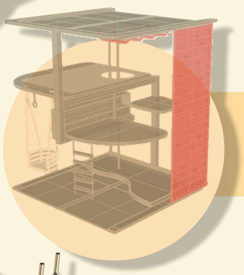
Customize via App
Easily adapt to different home types and budgets.



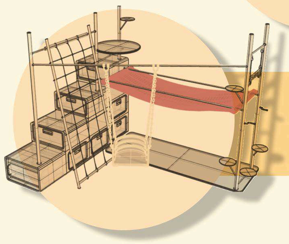
Self - assemble
Detachable design for easy transport, cleaning, and replacement.



Private Space
Semi-enclosed area that increases comfort and security.



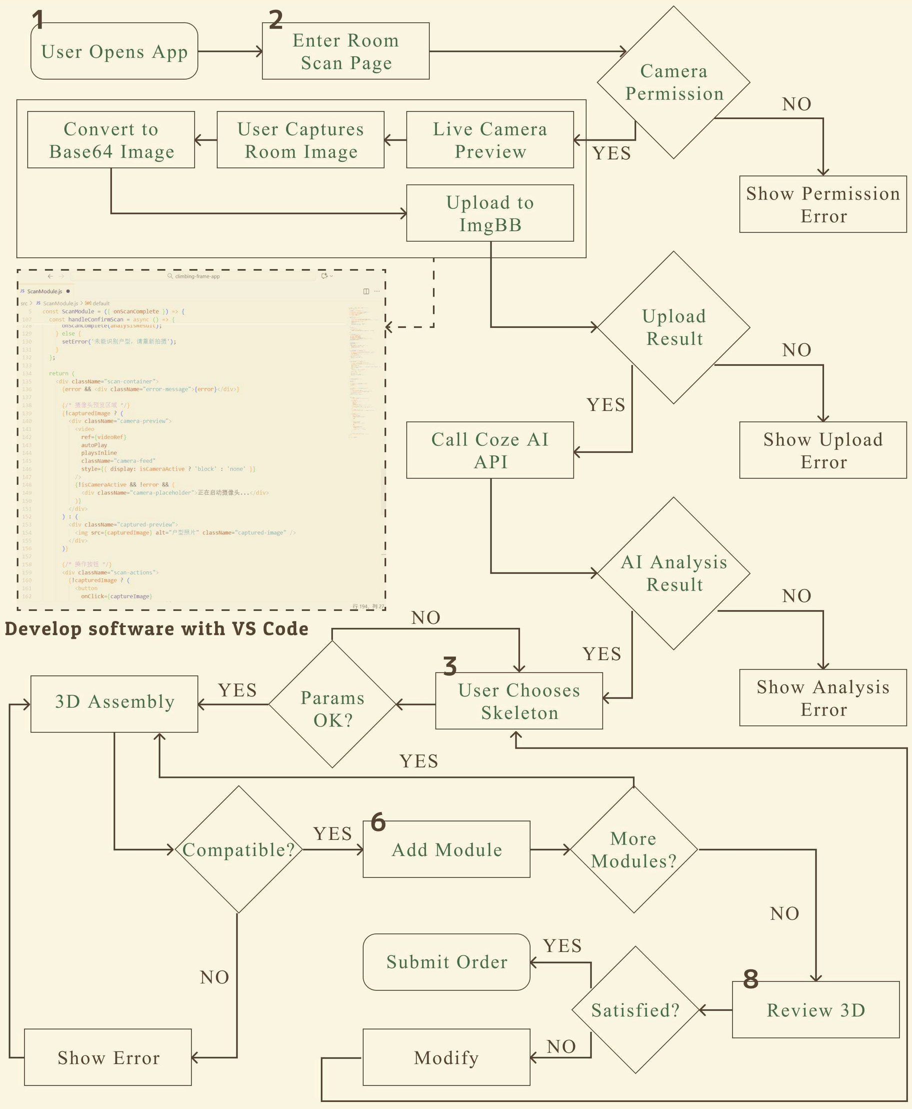
Return to Nature
Reawaken natural instincts through climbing and movement.



Rest Anytime
Like a childhood hideout — a cozy spot to relax after exercise.

INTERACTION PROCESS

INTERACTIVE FLOW CHART

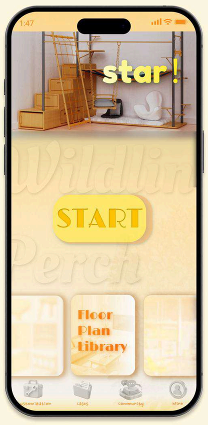

1
User Opens App
2
Enter Room Scan Page
Camera Permission
NO
Show Permission Error
YES
Live Camera Preview
User Captures Room Image
Convert to Base64 Image
Upload to ImgBB
Upload Result
NO
Show Upload Error
YES
Call Coze AI API
AI Analysis Result
NO
Show Analysis Error
YES
3
User Chooses Skeleton
Params OK?
NO
YES
3D Assembly
Compatible?
YES
6
Add Module
More Modules?
YES
NO
8
Review 3D
Satisfied?
YES
Submit Order
NO
Modify
NO
Show Error
climbing-frame-app
ScanModule.js
src > ScanModule.js > default
行 194，列 27

```
const ScanModule = ({ onScanComplete }) => {
  const handleConfirmScan = async () => {
        onScanComplete(analysisResult);
      } else {
        setError('未能识别户型，请重新拍摄');
      }
    };

    return (
      <div className="scan-container">
        {error && <div className="error-message">{error}</div>}

        {/* 摄像头预览区域 */}
        {!capturedImage ? (
          <div className="camera-preview">
            <video
              ref={videoRef}
              autoPlay
              playsInline
              className="camera-feed"
              style={{ display: isCameraActive ? 'block' : 'none' }}
            />
            {!isCameraActive && !error && (
              <div className="camera-placeholder">正在启动摄像头...</div>
            )}
          </div>
        ) : (
          <div className="captured-preview">
            <img src={capturedImage} alt="户型照片" className="captured-image" />
          </div>
        )}

        {/* 操作按钮 */}
        <div className="scan-actions">
          {!capturedImage ? (
            <button
              onClick={captureImage}
```

Develop software with VS Code

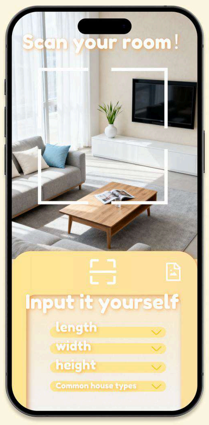
INTERACTIVE INTERFACE PRESENTATION

Start interface


1:47
star!
Wildlin
START
Perch
Floor Plan Library
customization
cases
community
Mine

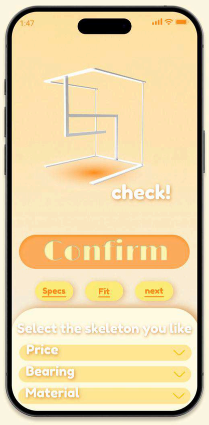
1

Identify the apartment layout


Scan your room!
Input it yourself
length
width
height
Common house types

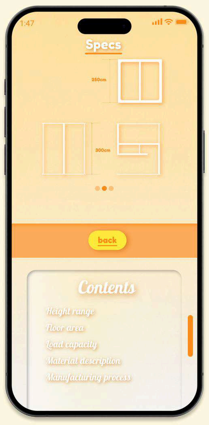
2

View the framework


1:47
check!
Confirm
Specs
Fit
next
Select the skeleton you like
Price
Bearing
Material

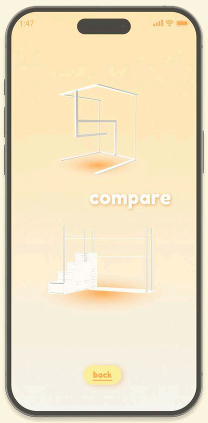
3

View the dimensions


1:47
Specs
250cm
300cm
back
Contents
Height range
Floor area
Load capacity
Material description
Manufacturing process

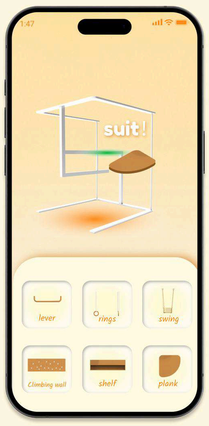
4

Compare the frameworks


1:47
compare
back

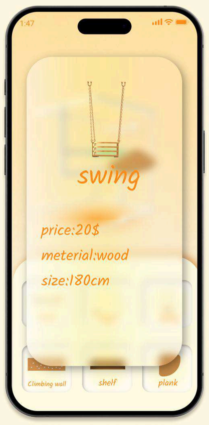
5

Add modules


1:47
suit!
lever
rings
swing
Climbing wall
shelf
plank

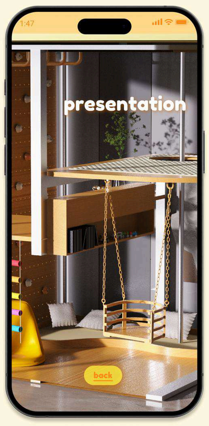
6

View the modules


1:47
swing
price:20$
meterial:wood
size:180cm
Climbing wall
shelf
plank

7

View the effect

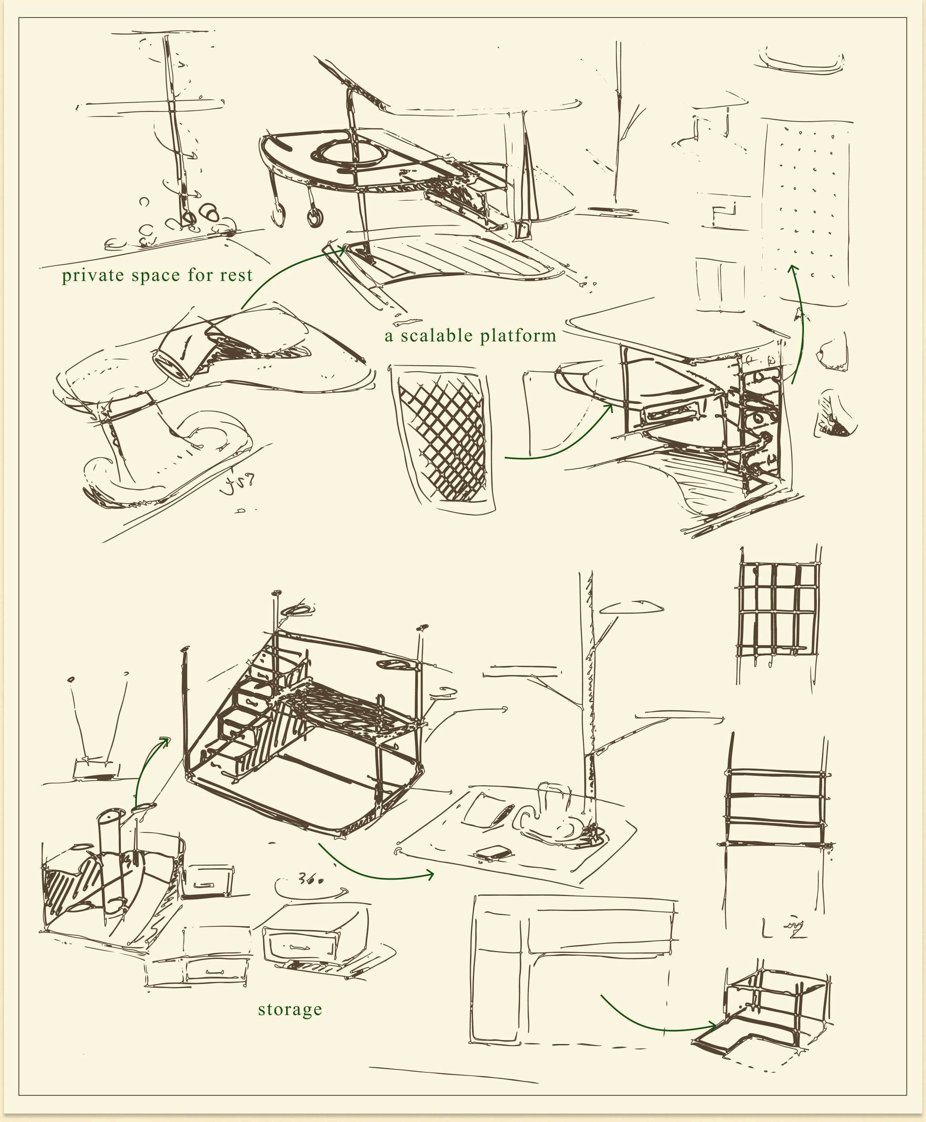

1:47
presentation
back

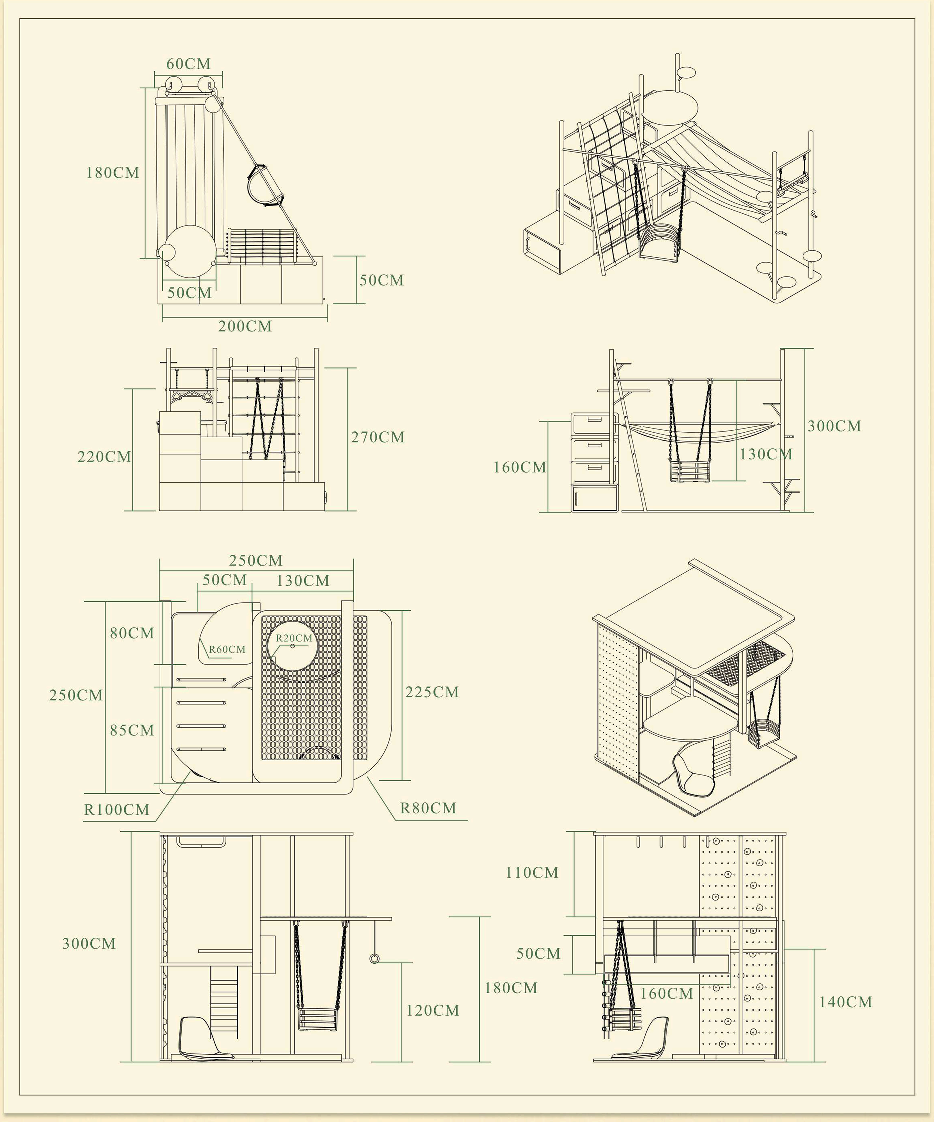
8

PRODUCT DEDUCTION

SKETCH DEPICTED IN VIEW OF THE ISSUES

DIMENSION ANALYSIS

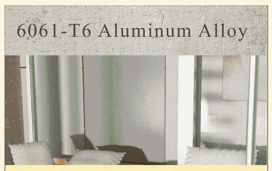

private space for rest
a scalable platform
storage


60CM
180CM
50CM
50CM
200CM
220CM
270CM
160CM
130CM
300CM
250CM
50CM
130CM
80CM
R60CM
R20CM
250CM
85CM
225CM
R100CM
R80CM
300CM
120CM
110CM
50CM
180CM
160CM
140CM

VIABILITY ASSESSMENT

MATERIAL & STRUCTURAL VERIFICATION

6061-T6 Aluminum Alloy



Aluminum Column
(Compression Stability)

$60\,\mathrm{mm} * 40\,\mathrm{mm} * 3\,\mathrm{mm}$

$Data : \sigma \approx 0.65\,\mathrm{MPa} < 398\,\mathrm{MPa}$

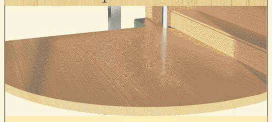
6061-T6 Aluminum Alloy

Aluminum Beam
(Bending Strength)

$50\,\mathrm{mm} * 30\,\mathrm{mm} * 2.5\,\mathrm{mm}$

$Data : \sigma \approx 40.1\,\mathrm{MPa} < 275\,\mathrm{MPa}$

Bamboo-Wood Composite Panel



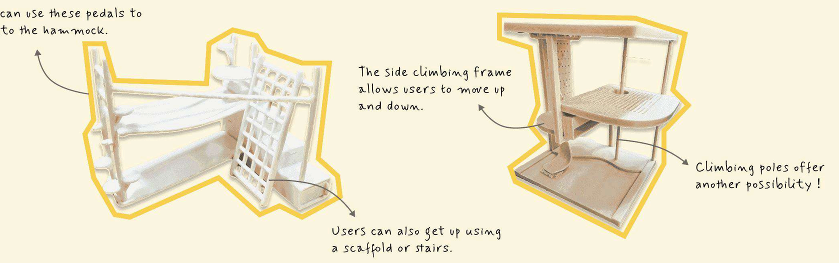
Bamboo-Wood Platform
(Bending Strength)

$Thickness: 25\,\mathrm{mm}$
$verification\ span: 1500\,\mathrm{mm}$

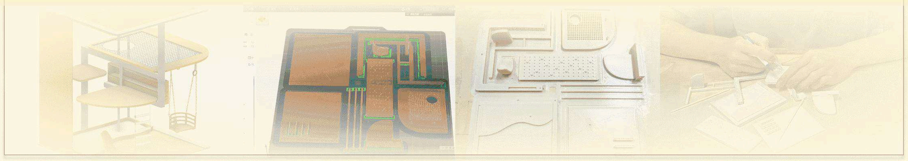
$Data : \sigma'_{\mathrm{wood}} \approx 1.32 MPa < 35 MPa$

Verification based on double load (2×75kg) for safety; all dimensions meet structural safety requirements.

3D - PRINT TO VALIDATE STRUCTURE


Users can use these pedals to get onto the hammock.
The side climbing frame allows users to move up and down.
Climbing poles offer another possibility !
Users can also get up using a scaffold or stairs.

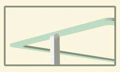
Modeling Slicing Printing Assembling



FEM ANALYSIS VALIDATES THE STABILITY

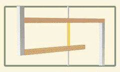
FEM analysis conducted to verify stability of a Frame-in-Frame structure.

global DOF constrained.

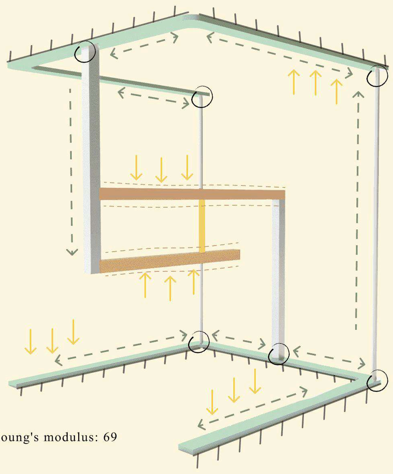


General constraints applied; top and base fixed to wall

bending member



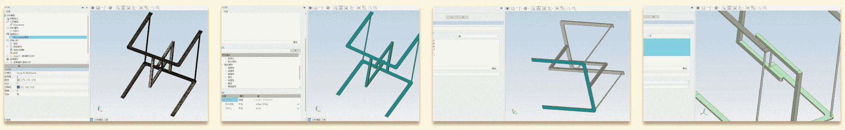
Pressure points selected; 4900 Pa load applied based on calculations.



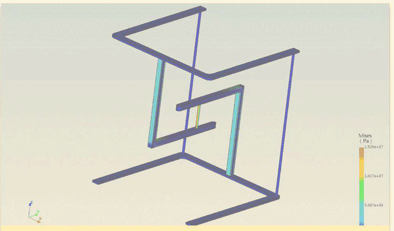
- 🟩 platform contact/user contact areas
- 🟨 added supportcomponents
- 🟫 bending/deformation components
- ↓ 2x75kg vertical load
- - - -> load transfer path
- /// wall-fixedconstraint
- - - - bending/displacement zone
- ○ Key joint/contact point

Specify material properties - Young's modulus: 69 GPa, Poisson's ratio: 0.33.

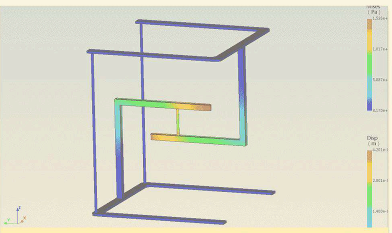
RESULTS
Max stress=15.26MPa<275MPa
Stress:PASS
Displacement:support needed
Added components improve local stability



Parameter settings are configured for the test model, including material, constraints, and stress surfaces.



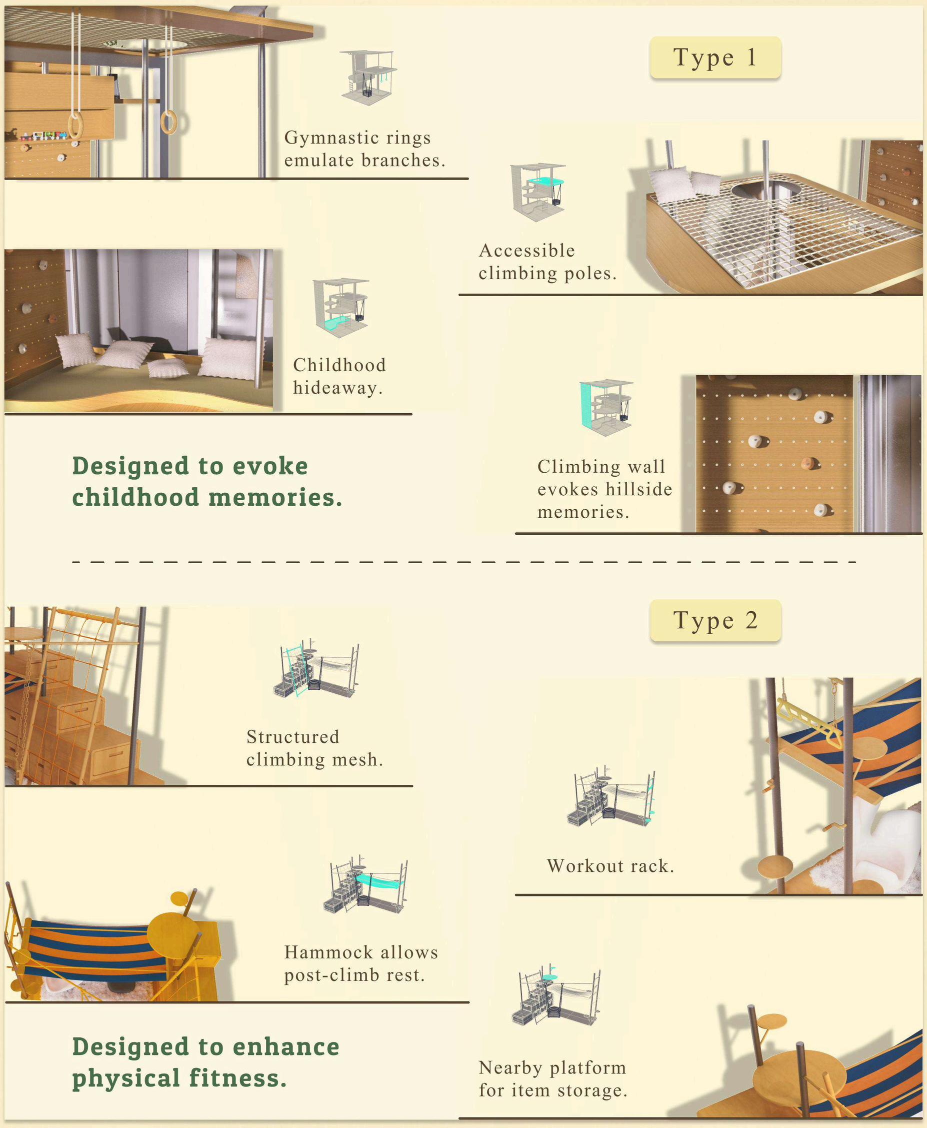
Von Mises Stress Plot



Displacement Plot

PRESENTATION

DETAILED PRESENTATION



Type 1

Gymnastic rings emulate branches.

Accessible climbing poles.

Childhood hideaway.

Climbing wall evokes hillside memories.

Designed to evoke childhood memories.

Type 2

Structured climbing mesh.

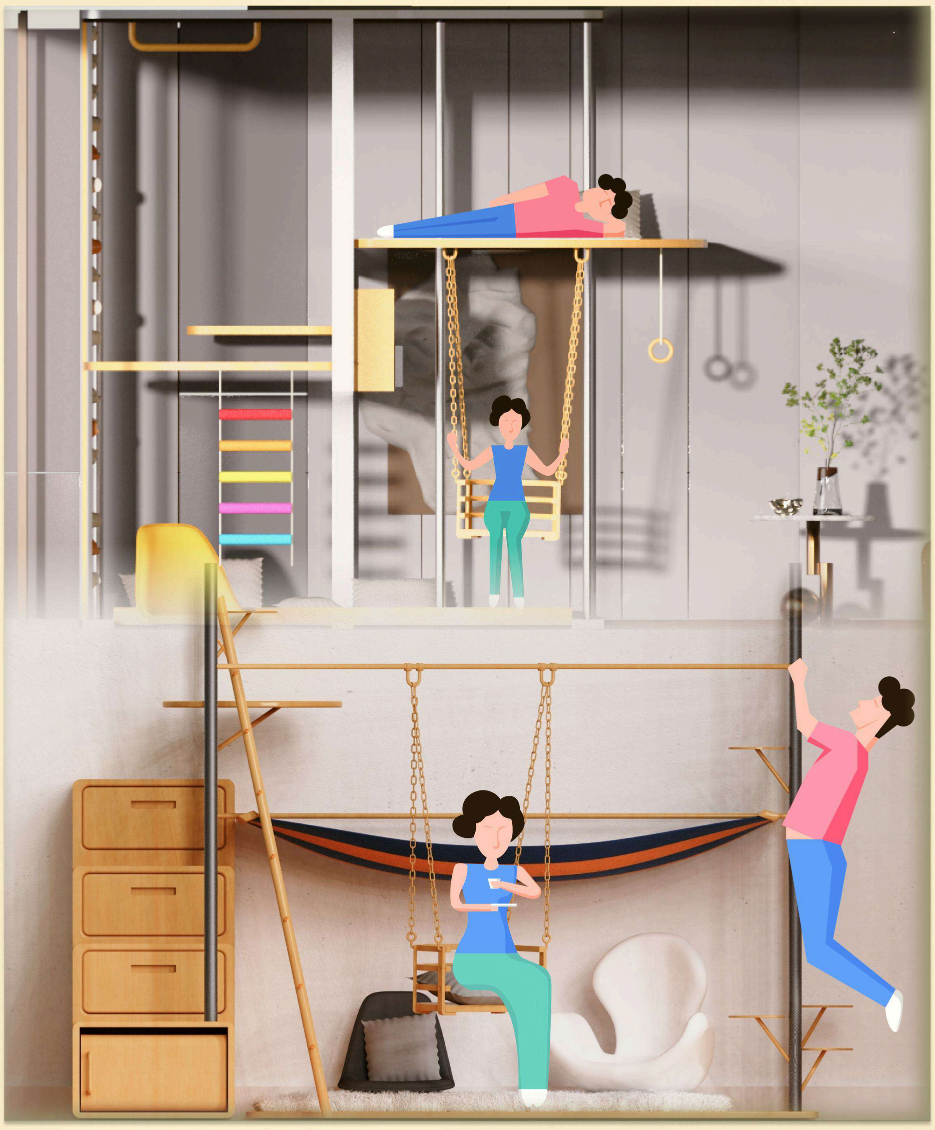
Workout rack.

Hammock allows post-climb rest.

Nearby platform for item storage.

Designed to enhance physical fitness.

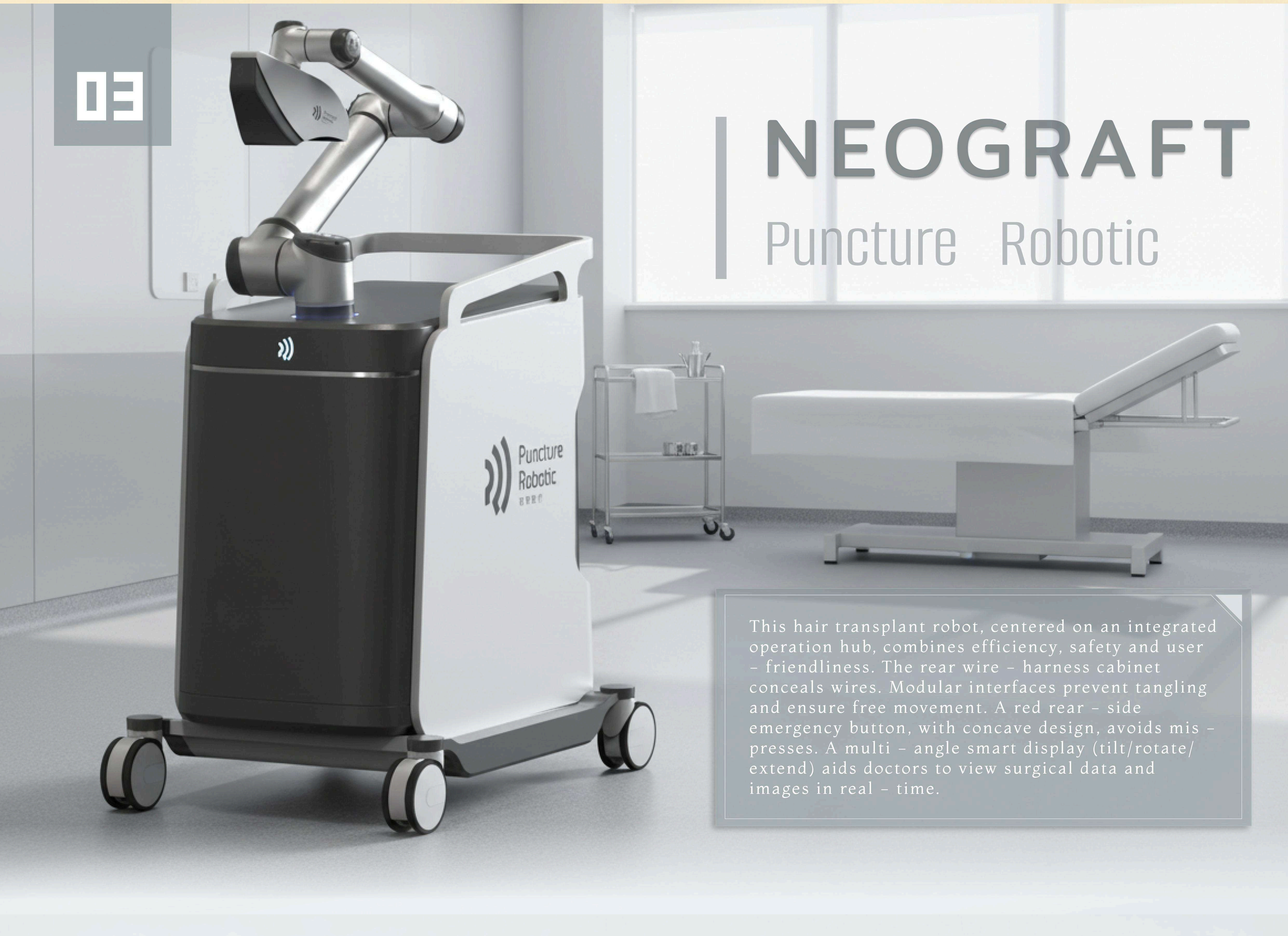
SCENARIO SIMULATION



03

NEOGRAFT

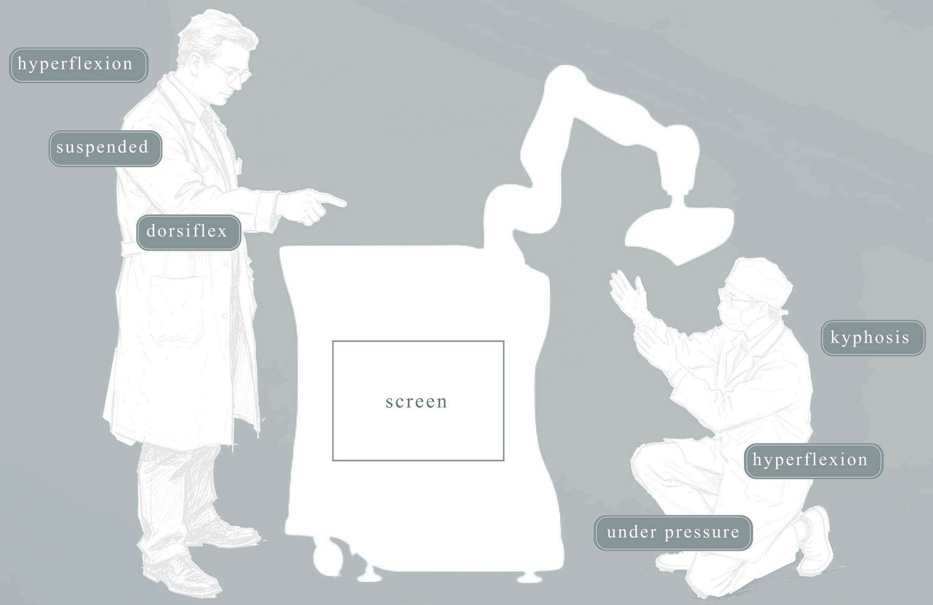
Puncture Robotic


Puncture
Robotic

This hair transplant robot, centered on an integrated operation hub, combines efficiency, safety and user - friendliness. The rear wire - harness cabinet conceals wires. Modular interfaces prevent tangling and ensure free movement. A red rear - side emergency button, with concave design, avoids mis - presses. A multi - angle smart display (tilt/rotate/ extend) aids doctors to view surgical data and images in real - time.

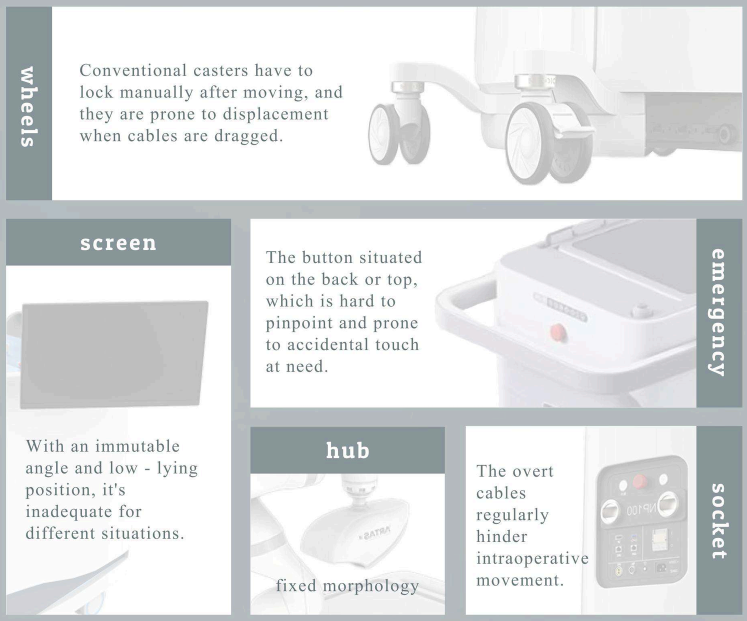
BACKGROUND

PIN POINT


hyperflexion
suspended
dorsiflex
screen
kyphosis
hyperflexion
under pressure

The neck, waist, knees and ankles may incur varying degrees of damage during the work. In view of safety, doctors' positions cannot be adjusted arbitrarily.

DESIGN POINT


wheels
screen
emergency
hub
socket

wheels

Conventional casters have to lock manually after moving, and they are prone to displacement when cables are dragged.

screen

With an immutable angle and low - lying position, it's inadequate for different situations.

emergency

The button situated on the back or top, which is hard to pinpoint and prone to accidental touch at need.

hub

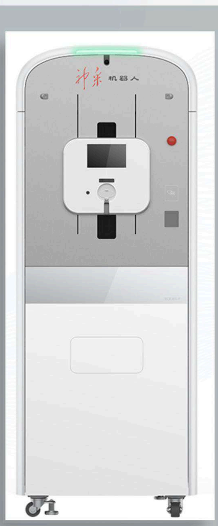
fixed morphology

socket

The overt cables regularly hinder intraoperative movement.

STYLE DEFINITION

INSPIRATION



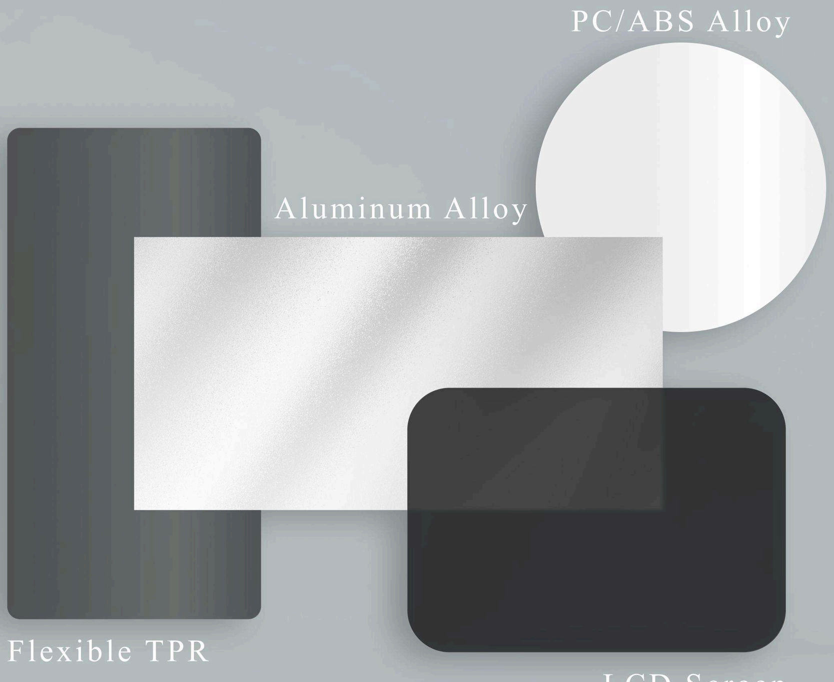
Minimalism

Metallic

Whiteness

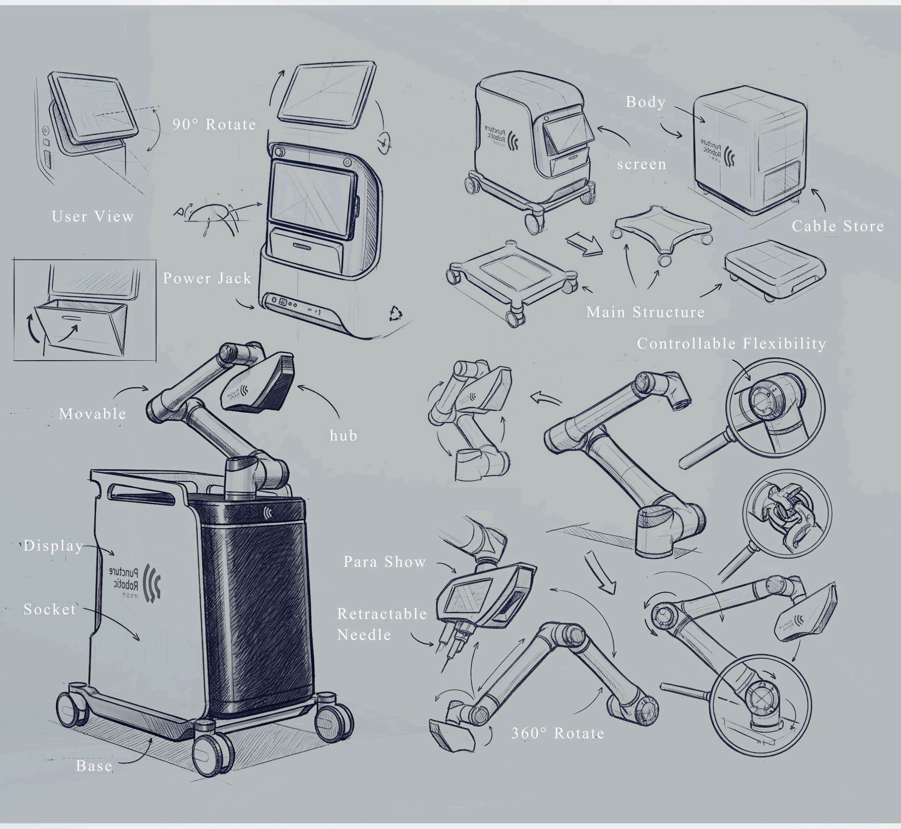
Sinuosity

MATERIAL &COLOR

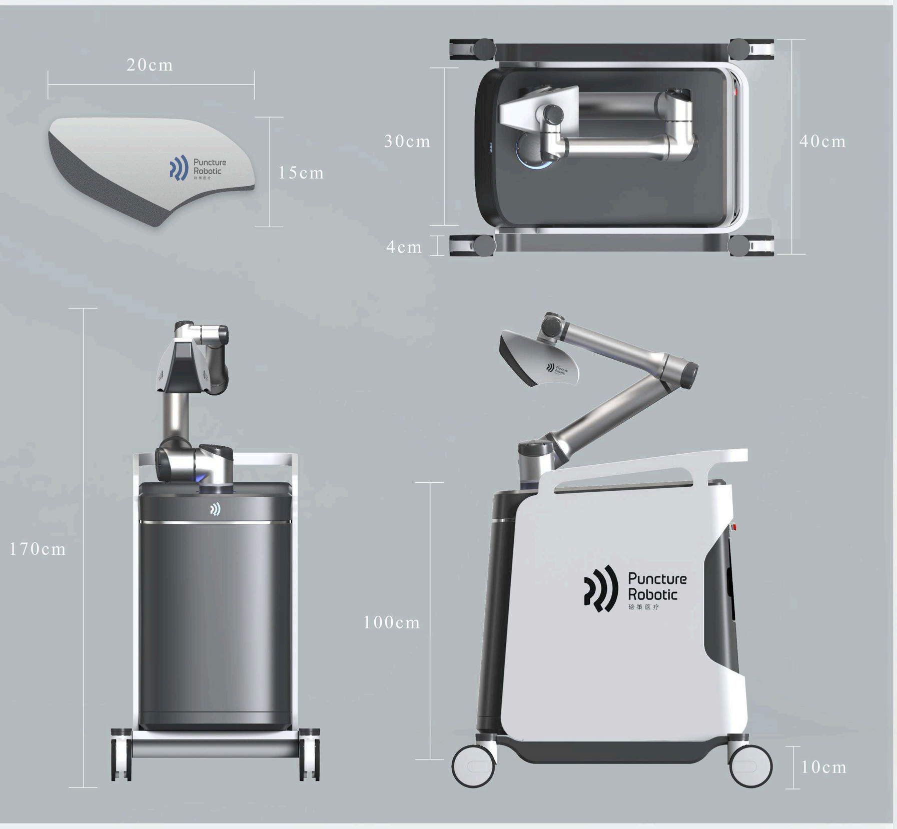

PC/ABS Alloy
Aluminum Alloy
Flexible TPR
LCD Screen

APPEARANCE CONCEPTION

SKETCH

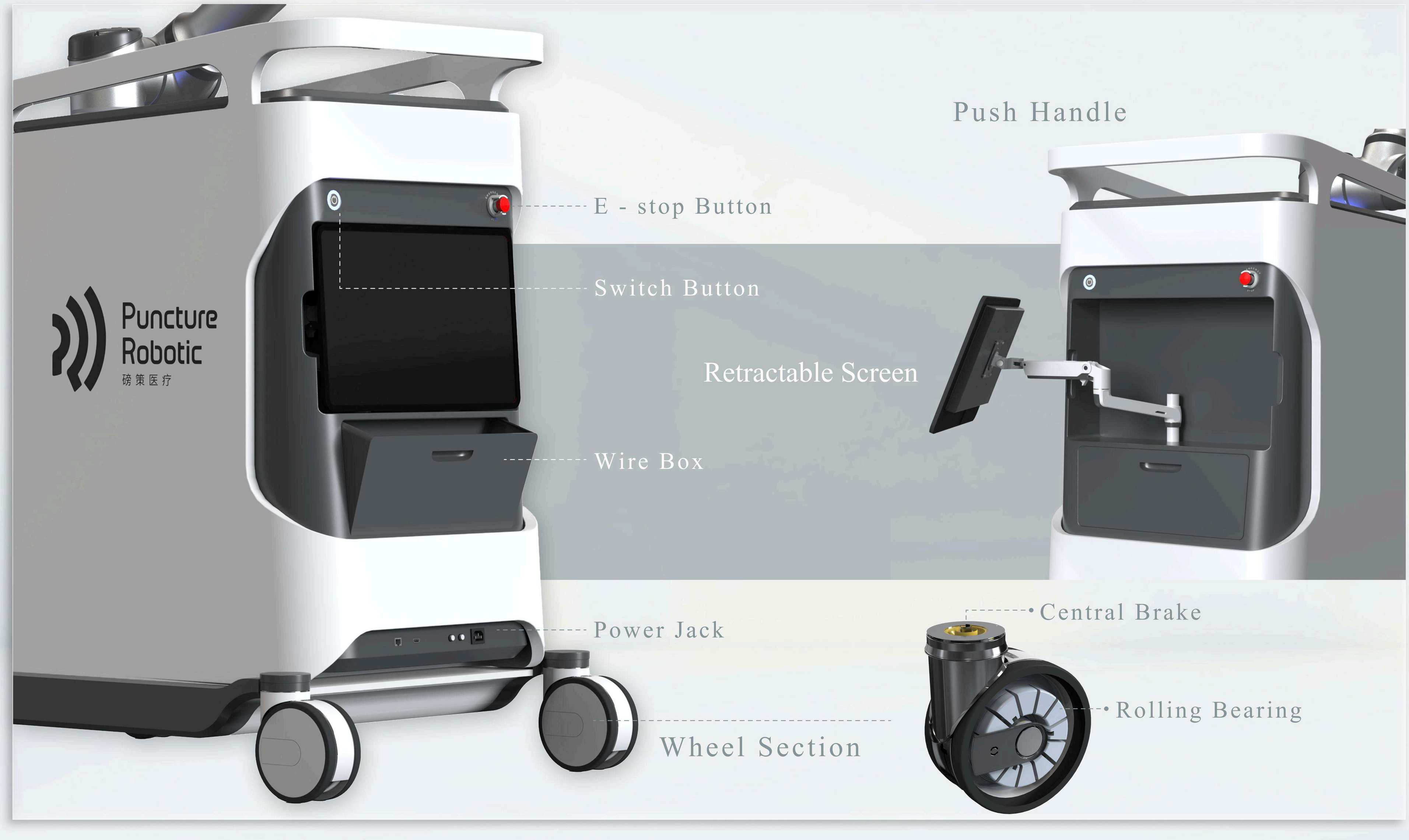

90° Rotate
User View
Power Jack
Body
screen
Cable Store
Main Structure
Controllable Flexibility
Movable
hub
Display
Socket
Para Show
Retractable Needle
360° Rotate
Base

DIMENSION


20cm
15cm
30cm
40cm
4cm
170cm
100cm
10cm
Puncture Robotic

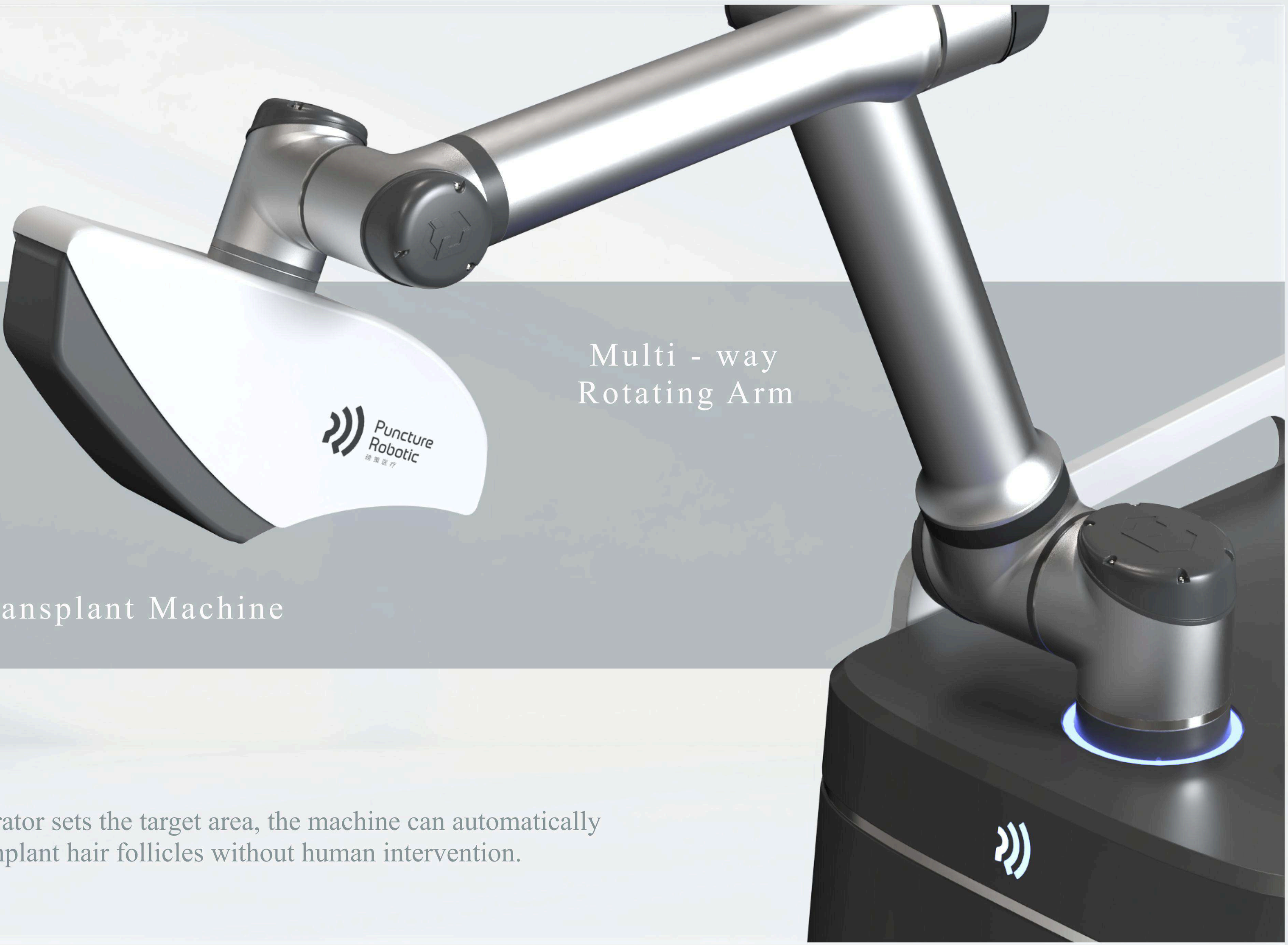
DETAIL DISPLAY

BODY DISPLAY


Push Handle
E - stop Button
Switch Button
Retractable Screen
Wire Box
Power Jack
Central Brake
Rolling Bearing
Wheel Section
Puncture Robotic
磅策医疗

DETAIL DISPLAY

ROBOTIC ARM DEMONSTRATION

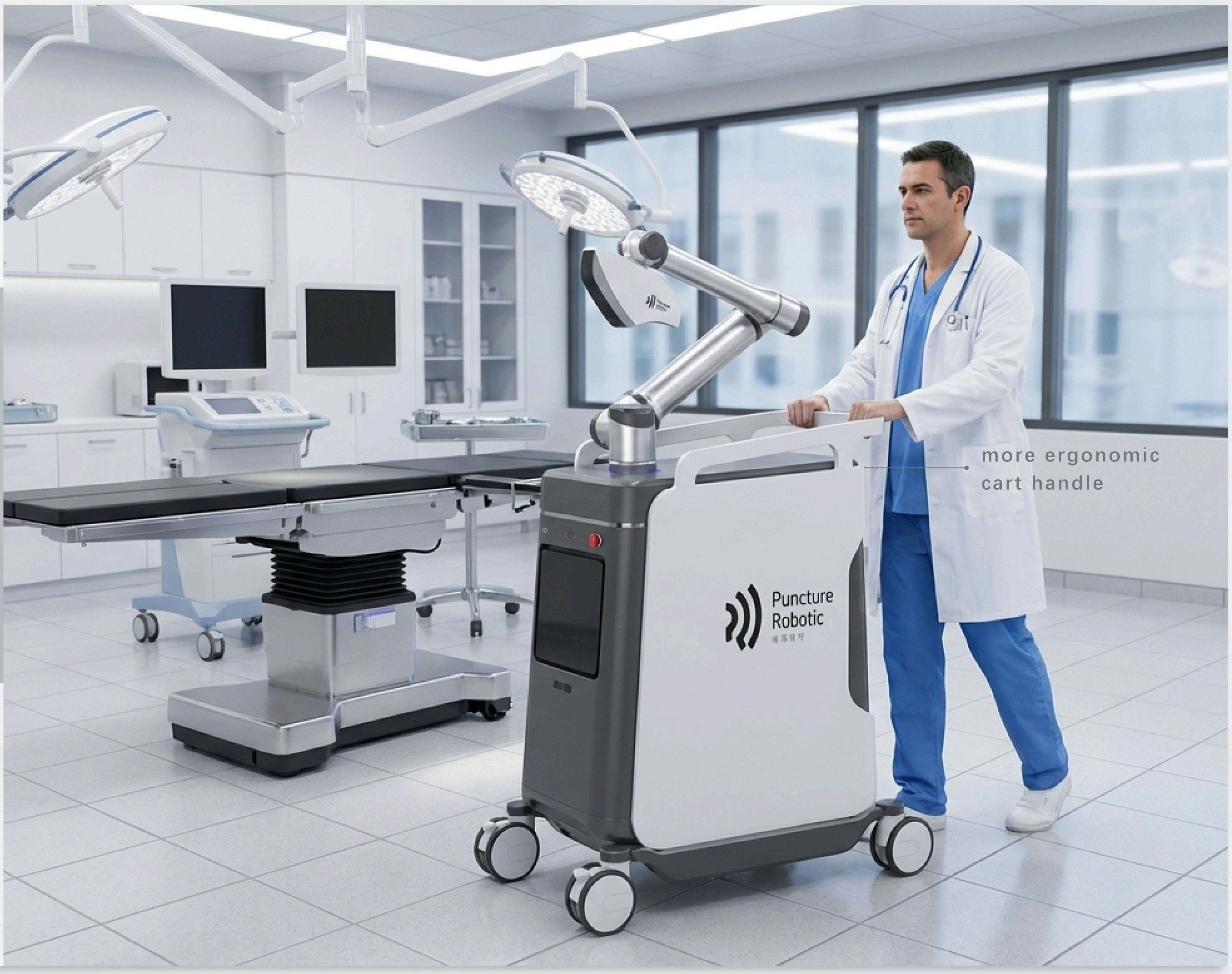

Multi - way Rotating Arm
Puncture Robotic
Hair - transplant Machine

After the operator sets the target area, the machine can automatically extract and implant hair follicles without human intervention.

SCENE DISPLAY

MOBILE PRESENTATION

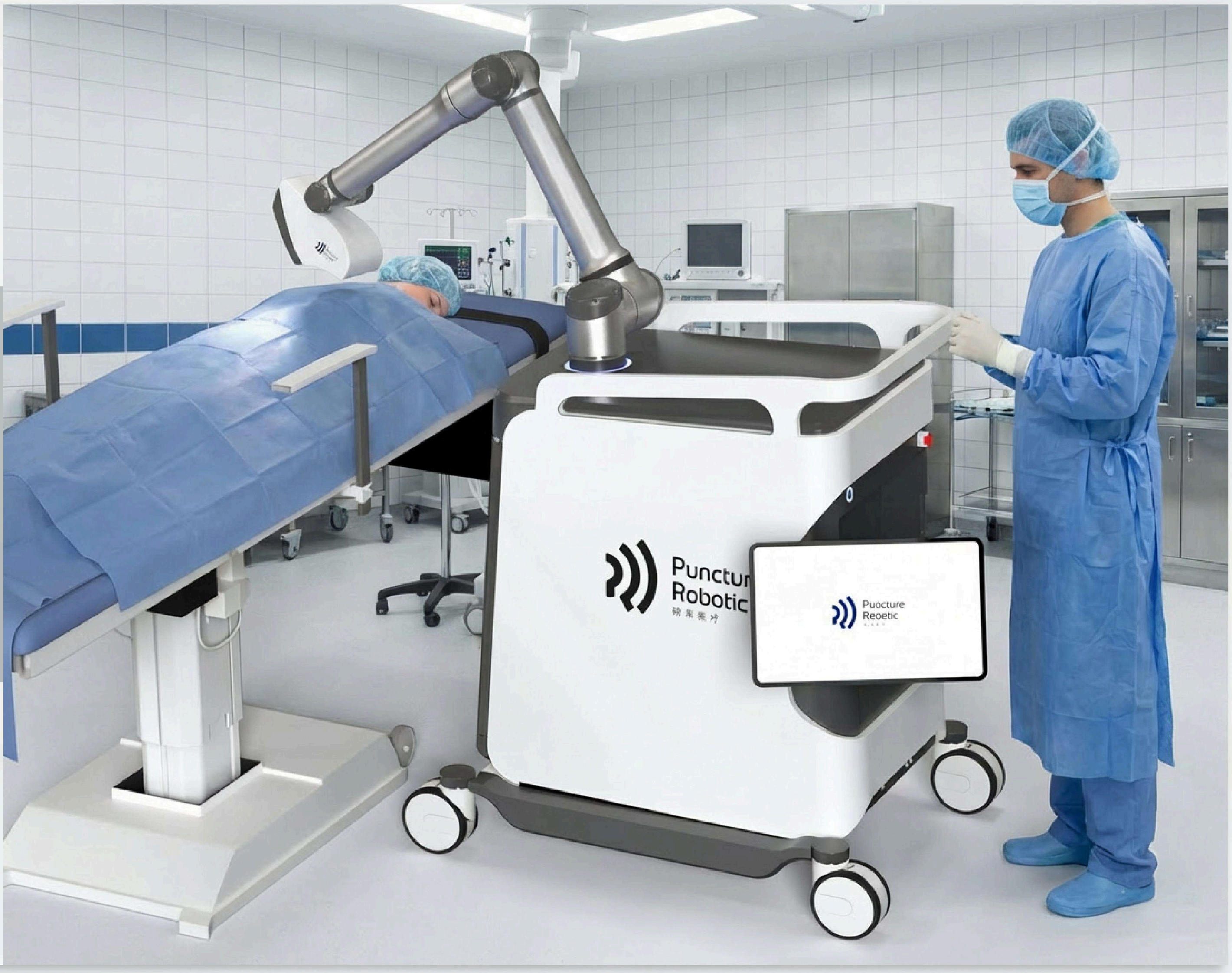
Four universal wheels suit narrow spaces/surgical instruments needing angle fine - tuning, fitting this product's needs.Central - control locking system: different locks control steering & forward movement. Rolling - bearing design cuts moving noise, suits medical environment.


Puncture Robotic
more ergonomic cart handle

SCENE DISPLAY

WORK PRESENTATION

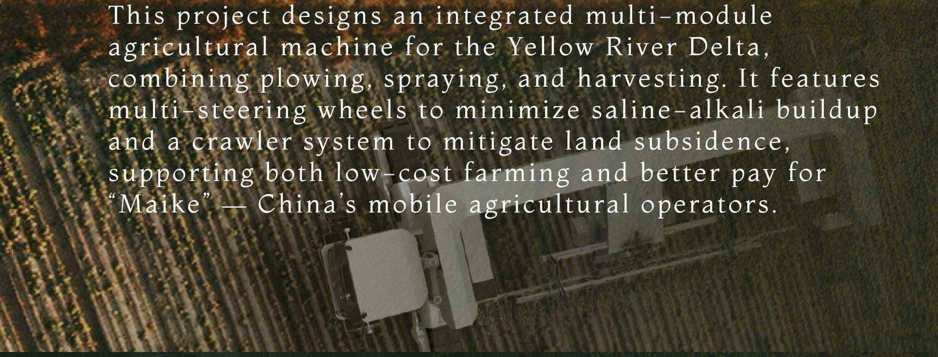
Prior to the operation, users adjust machine parameters via the display, which can move telescopically in all directions for doctors to operate from various angles. When idle, it can be stowed to prevent scratches and conserve space. In case of problems during the operation, press the emergency - stop button immediately.


Punctur
Robotic
Puocture
Reoetic

04

INTELLIGENT MULTI-MODULE AGRICULTURAL MACHINERY FOR THE YELLOW RIVER DELTA

GEOGRAPHIC OVERVIEW



This project designs an integrated multi-module agricultural machine for the Yellow River Delta, combining plowing, spraying, and harvesting. It features multi-steering wheels to minimize saline-alkali buildup and a crawler system to mitigate land subsidence, supporting both low-cost farming and better pay for “Maike” — China’s mobile agricultural operators.

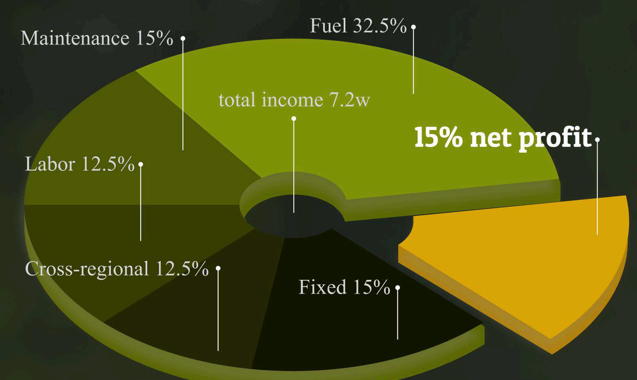
- 🟩 Severely Salinized Zones
- 🟨 Cultivated Land
- ▲ Wheat cultivation zone

In this area, due to river channel segmentation, cultivated land is fragmented. Large - scale machinery can hardly enter, and cross - plot transportation requires disassembly and manual handling.

Challenge

ISSUE DEMONSTRATION

INCOME AND EXPENDITURE ANALYSIS



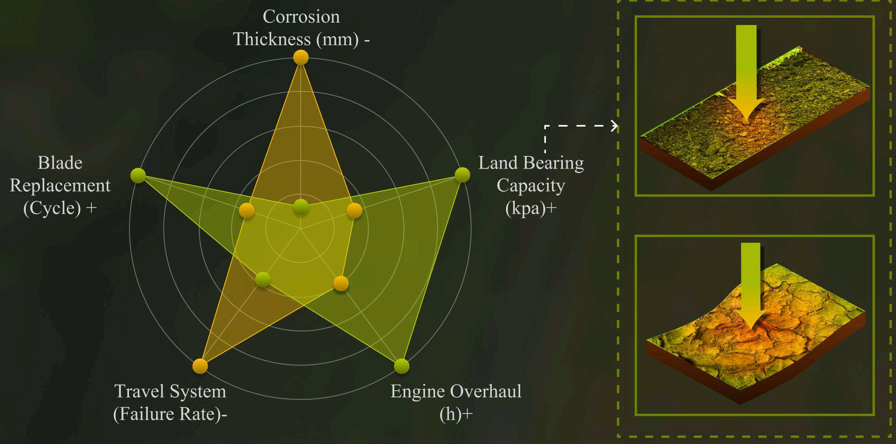
Challenge 1:
Efficiency in Operation

Challenge 2:
Durability and Maintenance

Challenge 3:
Energy Use Efficiency

Data from one wheat harvest using a combine harvester.

SOIL SALINIZATION ISSUE



● Typical soil ● Salinized soil +: Larger value = Better -: Smaller value = Better

Soil salinization has seriously affected the revenue and expenditure of agricultural work.

USER CHARACTERISTICS

PERSONA

Hopes
- economical
- streamlined
- productive

Maike *(individual farming in subsided areas)*

Age: 40
Experience: 8 years | Manages 8–10 mu

“The operation should be simple. One person should manage fuel efficiency and stability easily.”

TRAPPED BY EFFICIENCY

Hopes
- convenient
- durable
- rugged

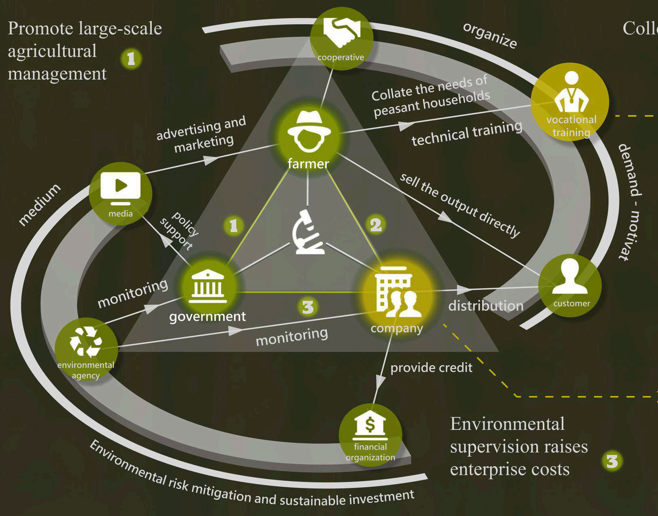
Farmer *(who possess a few pieces of land)*

Age: 45
Experience: 15 years | Manages 15 mu

“The machine must be reliable and adapt to rugged, saline-alkali terrain.”

TRAPPED BY ENVIRONMENT

STAKEHOLDER


cooperative
organize
Collate the needs of peasant households
vocational training
technical training
advertising and marketing
farmer
media
medium
policy support
1
2
sell the output directly
demand - motivat
customer
monitoring
government
3
company
distribution
monitoring
environmental agency
provide credit
financial organization
Environmental risk mitigation and sustainable investment

Promote large-scale agricultural management ①

Collect farmers’ needs ②

Environmental supervision raises enterprise costs ③

Simplified operation reduces training costs.

Integrated functions reduce production costs.

DESIGN CONCEPTION

THE INSPIRATION FROM CRABS

Ecdysis——
Module Replacing

Segmented Exoskeleton ——
multi - modular assembly

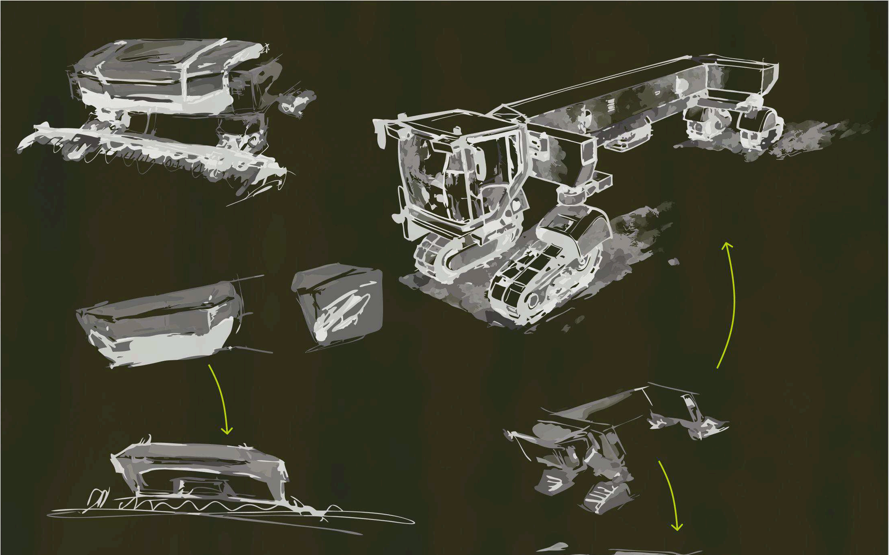
Laterigrade——
Multi - directional Movement

STUDY MODEL WITH BLOCKS

Model assembly **Forward mode**

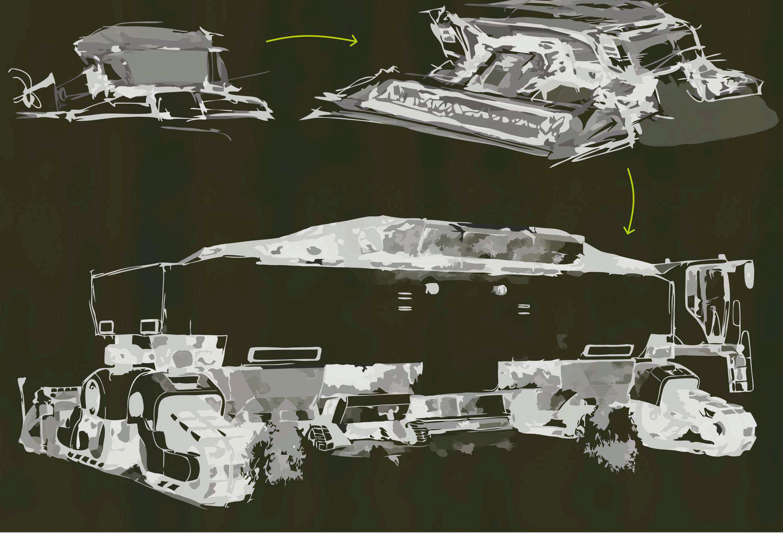
SKETCH

WHAT SHOULD OUR PRODUCT BE LIKE?



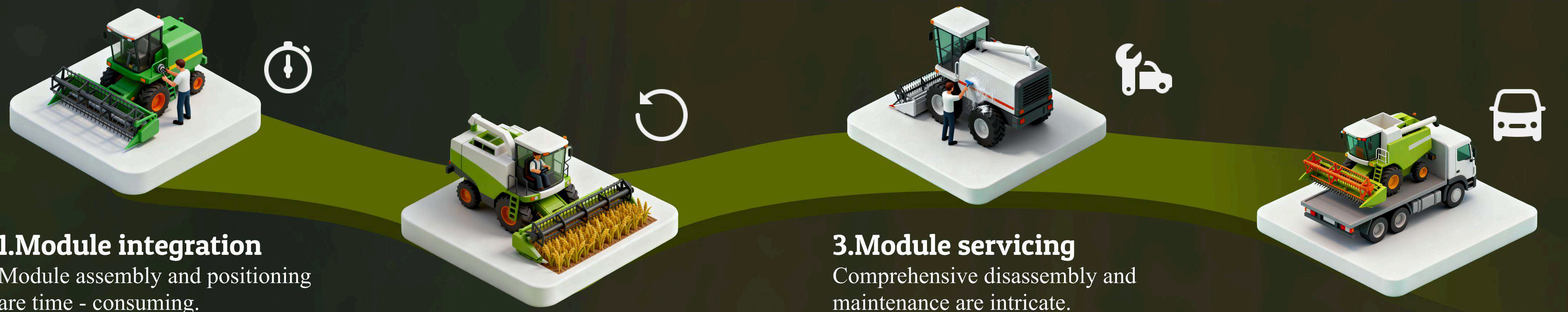
Sideways movement mode

Spin - on - the - spot mode



DESIGN ORIENTATION

OPERATIONAL PROCEDURES AND ISSUE RECAPITULATION

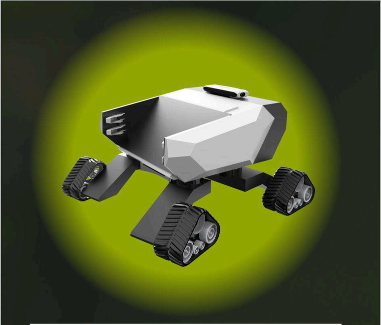


1.Module integration
Module assembly and positioning are time - consuming.

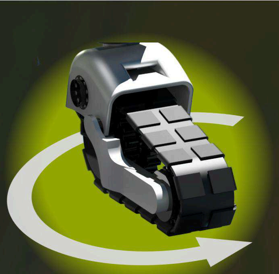
2.Field work
There are numerous turning blind spots. It is liable to induce vehicle bogging.

3.Module servicing
Comprehensive disassembly and maintenance are intricate.

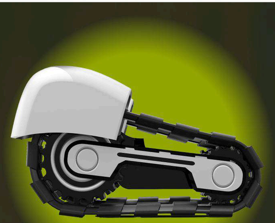
4.Interregional transportation
The transportation cost of the entire unit is high.



Automatic assembly
employ an autonomous cart for modular integration to boost efficiency.



Crawler wheel
By increasing the contact area with the ground to mitigate the problem of vehicle bogging.



Swivel wheel
Minimize turning blind spots while diminishing the crevices where soil accumulates.

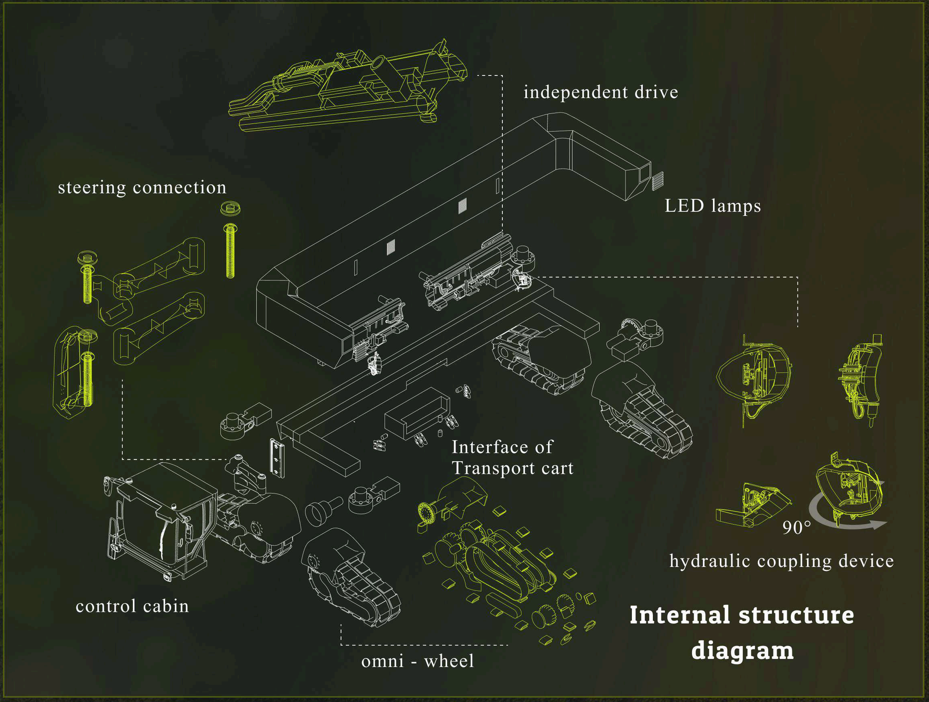
Modular integration
ntity to develop a multi - functional agricultural implement, thereby reducing the parts to be disassembled and transported.

MAIN STRUCTURE

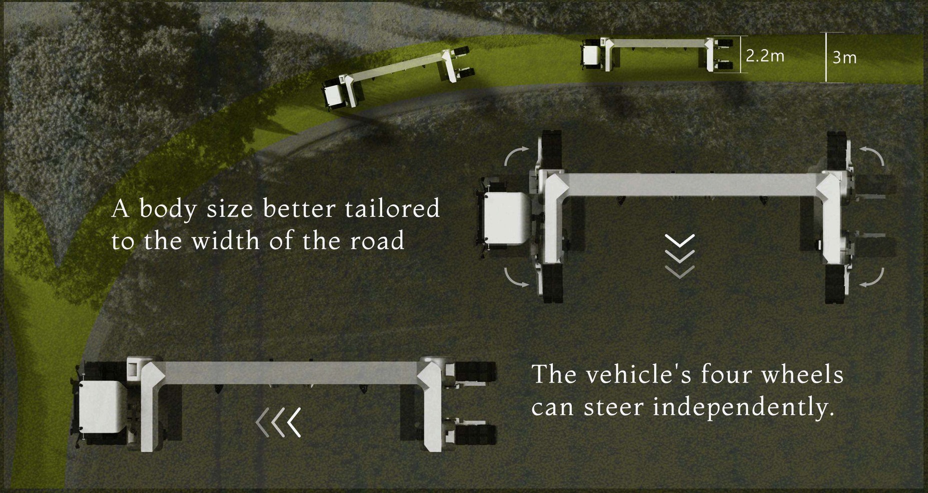
CORE STRUCTURE OF AGRICULTURAL MACHINERY

FUNCTIONAL ANALYSIS

CORE STRUCTURE'S MOVEMENT SHOWCASE


independent drive
steering connection
LED lamps
Interface of
Transport cart
90°
hydraulic coupling device
control cabin
omni - wheel

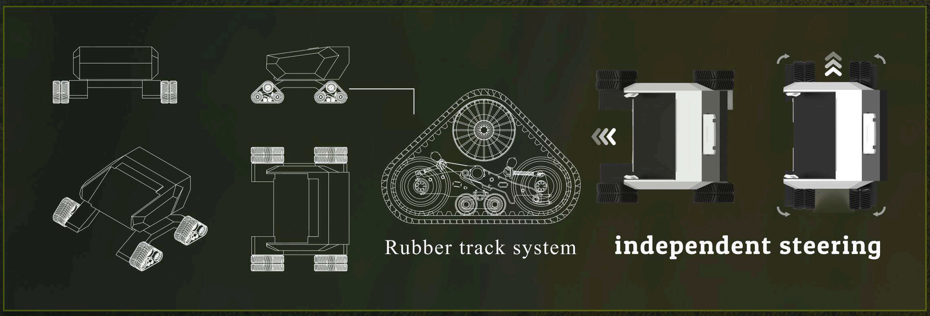
Internal structure diagram



A body size better tailored to the width of the road

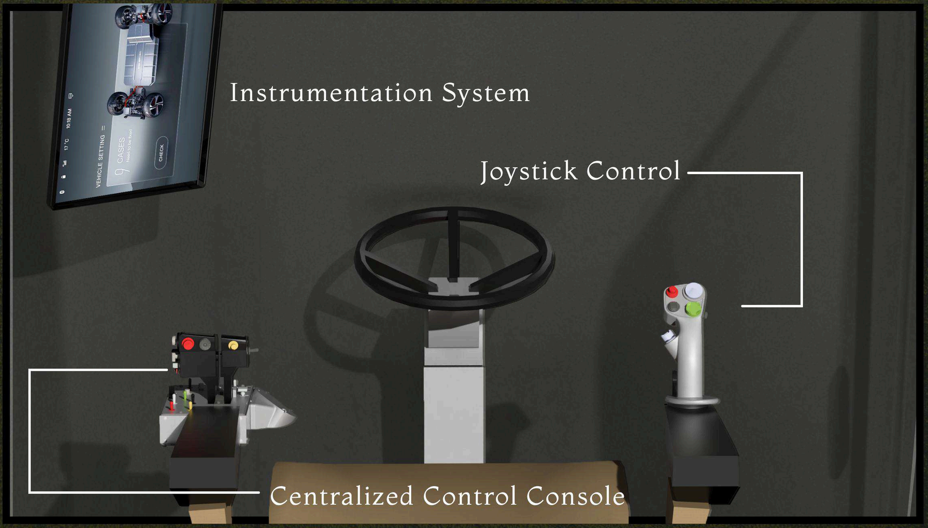
The vehicle's four wheels can steer independently.

TRANSPORT CART(AUTOMATIC LOADING OPERATING MODULE)


Rubber track system

independent steering

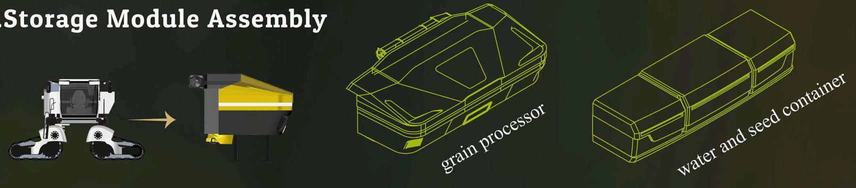
CAB INTERIOR STRUCTURE


Instrumentation System
Joystick Control
Centralized Control Console

COMPONENT CONNECTION

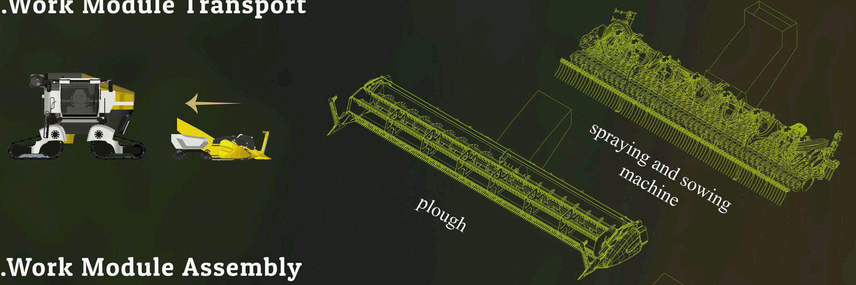
THE ASSEMBLY PROCESS

1.Storage Module Assembly

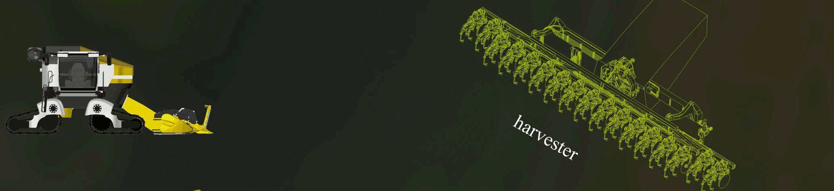

grain processor
water and seed container

storage module

2.Work Module Transport


plough
spraying and sowing
machine

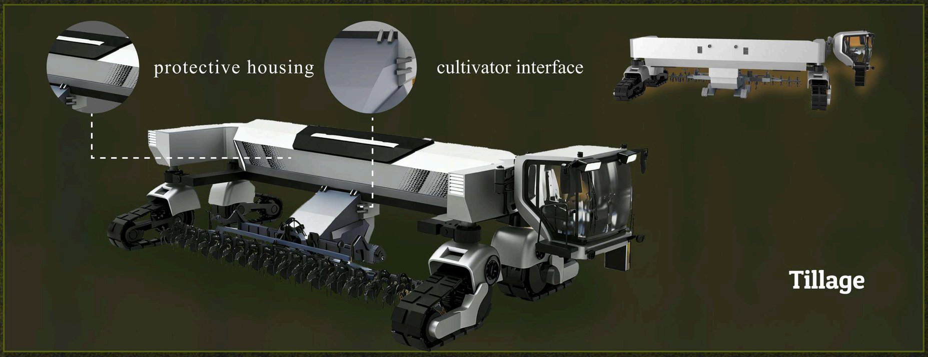
3.Work Module Assembly



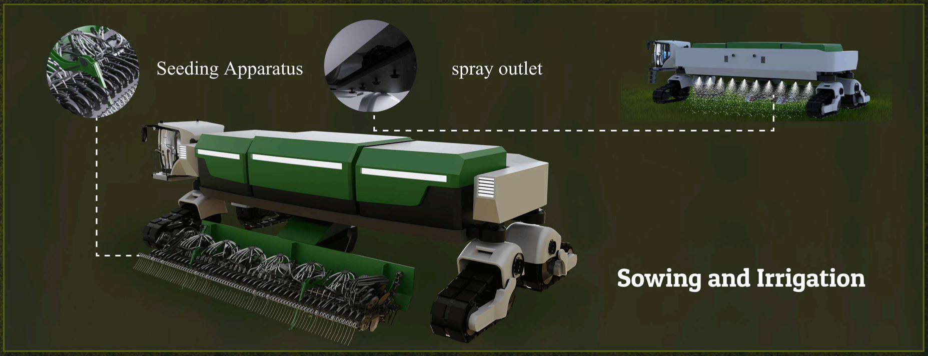
operating module

transport process

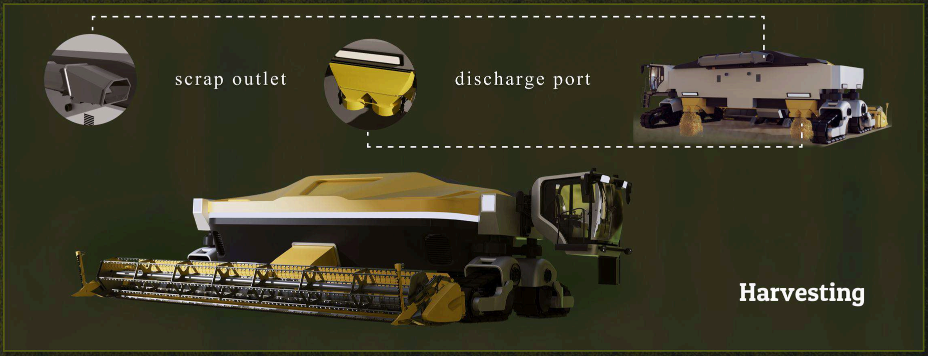
CONNECTION ACHIEVEMENT PRESENTATION


protective housing
cultivator interface

Tillage


Seeding Apparatus
spray outlet

Sowing and Irrigation


scrap outlet
discharge port

Harvesting